

Working with Objects in the Frame

Work with Objects

Plant Simulation provides a number of ways for working with the objects in the scene.

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Insert Objects into the Scene

You can insert objects into the 3D scene from the *Toolbox* or the *Class Library*. They contain the same objects by default.

Proceed as follows:

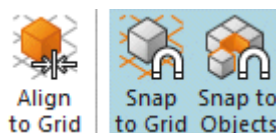
- To insert an object from the **Toolbox**, click the object you want to insert with the left mouse button and drag the mouse to the position in the 3D window at which you want to place the object. Click the mouse button to insert the object. Cancel inserting by clicking the right mouse button or by pressing **Esc**.
- To insert an object from the **Class Library**, click the object you want to insert with the left mouse button, hold down the mouse button and drag the mouse to the position in the 3D window at which you want to place the object. Then, release the mouse button. As long as you do not release the mouse button, *Plant Simulation* shows a preview of the object in the 3D window. You can cancel inserting by releasing the mouse outside of the 3D window.
- To insert an object which you copied with **Home > Copy** or with **Ctrl+C** to the *Clipboard*, press **Ctrl+V** in the 3D window to paste the copy into the object opened in the 3D window. *Plant Simulation* pastes the copy at the same position which the copied object takes within its location.

To insert an object from the *Toolbox*:

- If you want the object to snap to the closest intersection of two grid lines of where you click the mouse

button, click  on the **Edit** ribbon tab.

If you want to place the object freely, do not click . *Plant Simulation* snaps the object according to the settings for aligning to the grid or for aligning the object to the grid and/or to objects.



Note

When you insert **Markers** into your model, make sure that the *Markers* are **aligned**. This prevents *Plant Simulation* from computing unnecessary roundings between the *Markers*. Misaligned *Markers* might occur if you inserted *Markers* into your model while **Show Grid** was off.

- Click the object you want to insert with the left mouse button to select it. Then, drag the mouse to the location where you want to place it and release and click the mouse button.
- If need be, you can fine-tune the location of the object in the dialog **3D Properties** that clicking  on the **Home** ribbon tab opens.
- To open the dialog of the simulation object, click the button in the bottom left corner in the dialog **3D Properties** or double-click the object in the 3D scene. Modify settings here.

To insert the same object from the *Toolbox* into a *Frame* several times, one after the other without interruption, hold down **Ctrl** and select the object. The object remains selected in the *Toolbox* after inserting it and you can thus insert several of these objects one after the other. You can exit this mode by clicking the right mouse button or by pressing **Esc**.

After you inserted objects into your simulation model, you can **align them to the grid** , **make them snap to the grid** , **make them snap to other objects** , or **arrange them on the grid**. To be able to do so, make sure that **Show Grid**  **Grid** is on.

See also

[Move an Object With the Mouse](#)

[Work with the Grid](#)

[Back to Work with Objects](#)

Select Objects

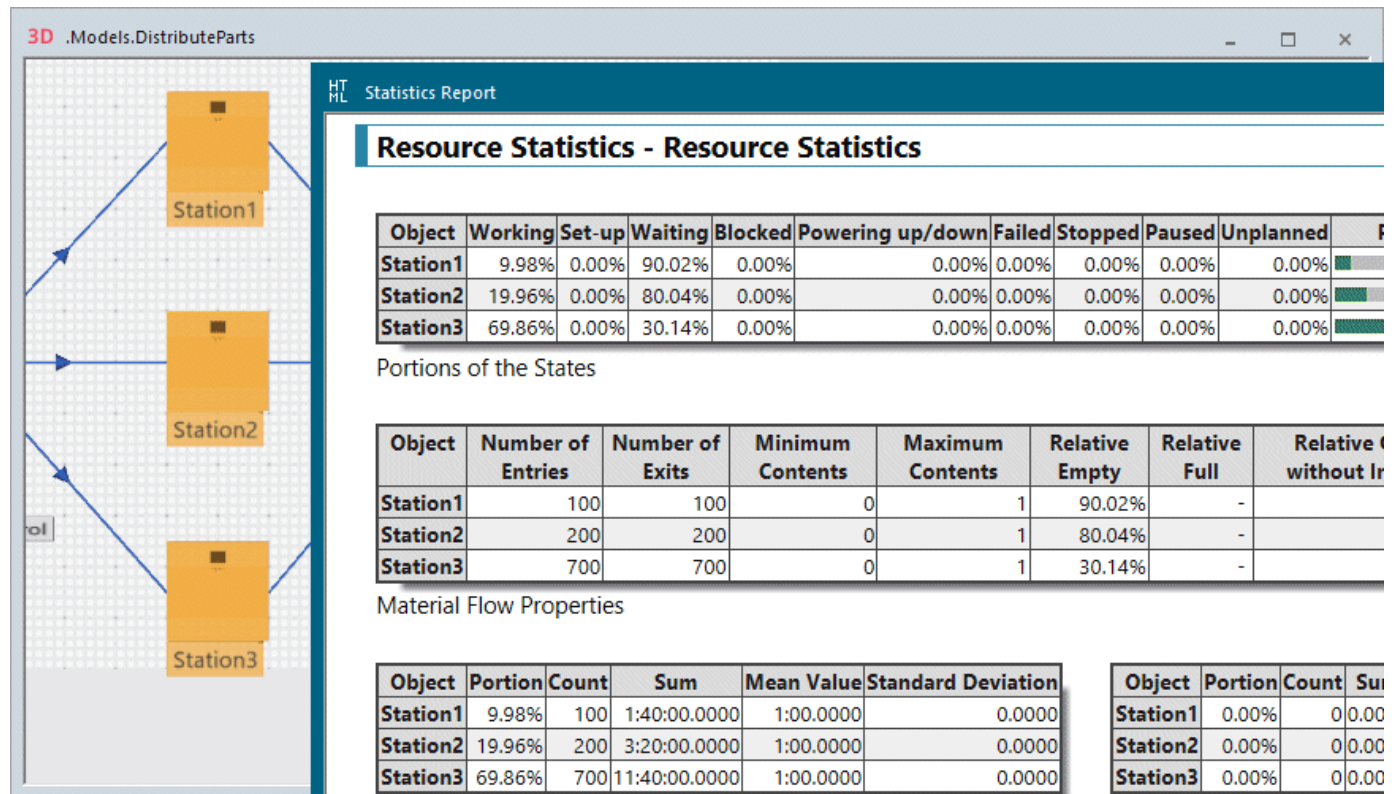
You can select objects in the 3D window by clicking it with the left mouse button.

You can:

- Select a **simulation object**, which is directly located in the object that is opened in the window.
Plant Simulation shows selected **simulation objects** in **green**.
- Select an **animatable object**, which belongs to the simulation object that is opened in the window.
Plant Simulation shows selected **animatable objects** in **purple**.
- Select a **graphic**, which visualizes the simulation object that is opened in the window.
Plant Simulation shows selected **graphics** in **yellow-green**.
- Select a displayed **manipulator**.
- Select two or more **items** by holding down the **Shift** or the **Ctrl** key and clicking the desired elements with the left mouse button.
- Select **several items** by dragging a marquee around the desired objects:
 - When you do not hold down the **Shift** or the **Ctrl** key, the selection with a marquee cancels an existing selection.
 - When you hold down the **Shift** or the **Ctrl** key, the new selection will be added to the existing selection.
- Press **Ctrl+A** to select all objects in the scene.

You can modify the selection with the plus-key (+) and the minus key (-), compare the [Quick Reference Card](#) > **While Modeling**.

To show objects, for example in the *statistics report*, in the sequence in which you selected them, hold down **Shift** and click the objects one after the other. In the example below we first clicked *Station1*, then *Station2*, and then *Station3*.



Compare [Move an Object With the Mouse](#)

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Simultaneously Paste Multiple Copies

At times you may want to paste more than one copy of the object you copied. You might, for example, want to create a *Store* with a number of shelves.

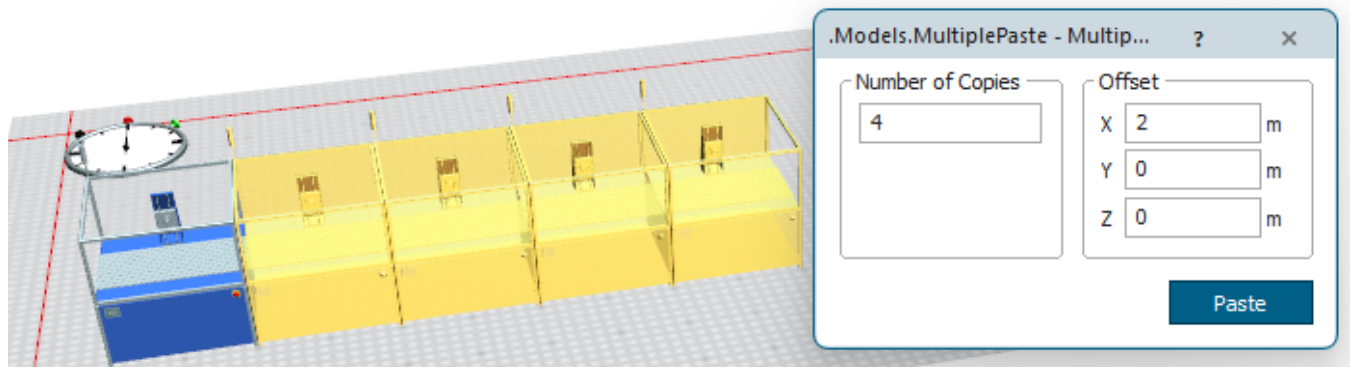
To paste multiple copies of an object:

- Insert the object you need, for example a *Station*, into the scene.
- Select the *Station* and copy it by pressing **Ctrl+C**.
- Click **Paste** > **Multiple Paste** on the **Home** ribbon tab.
- Enter the number of *Stations* you would like to paste into the combo box **Count**. We entered 4.

- Enter the **Offset** of the copies from the object you originally placed. Our *Station* is 2.0 meters wide:

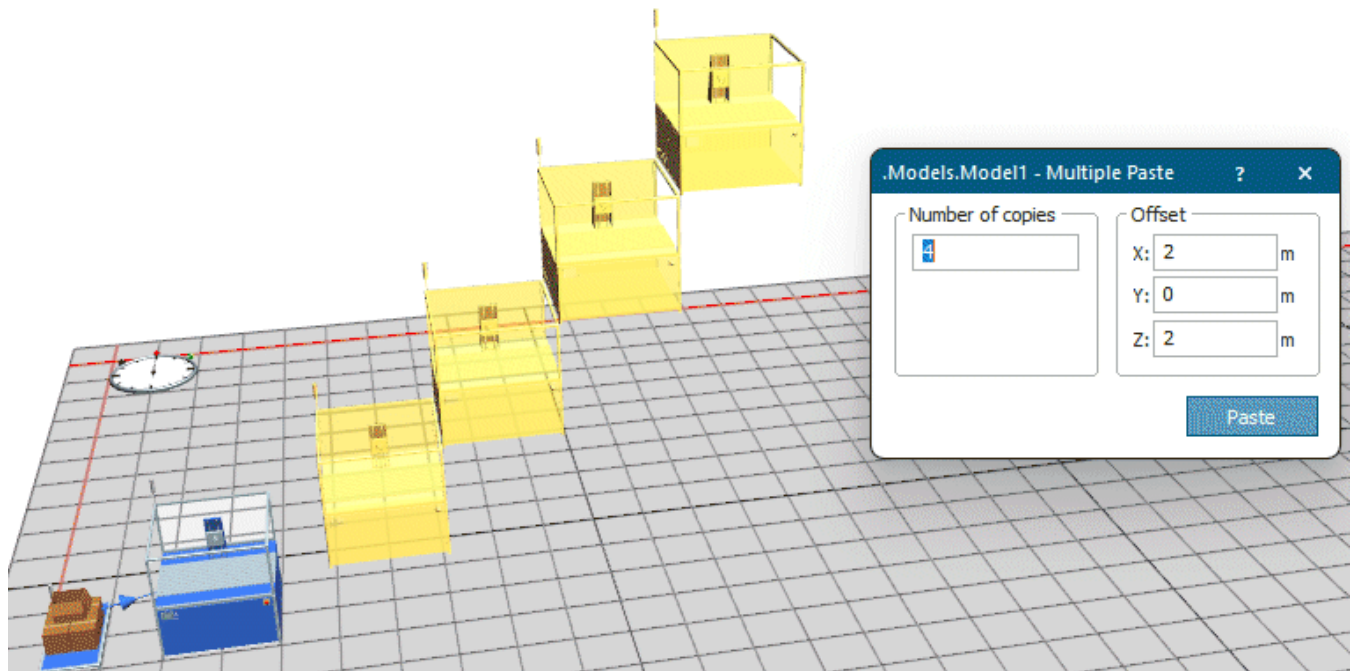
An offset of 2.0 meters on the **X-axis** pastes 4 copies of the *Station* with this **Offset** to the original toward the right. Note the initial predefined offset toward the right and downwards. We implemented this to prevent *Plant Simulation* from pasting the copies one on top of the other.

Click **Paste**. If need be, move the pasted copies to where you need them.



A **Number of Copies** of 4 and an **Offset** of 2.0 meters on the **X-axis** and on the **Z-axis** pastes 4 copies to the right and up of the original *Station*.

Click **Paste**. If need be, move the pasted copies to where you need them.



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Move Objects in the Frame

Move Objects in the Frame

Plant Simulation provides several ways to move an object in the scene.

You can:

- [Move an Object With the Mouse](#)
- [Move an Object With Keyboard Keys](#)
- [Move an Object Precisely](#)

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Move an Object With the Mouse

You can manipulate an object by clicking it with the left mouse button and dragging the mouse.

When you show the grid, manipulating the object relates to the grid plane. When you hide the grid, manipulating relates to the view plane.

- To **move the object freely on the grid plane**, click the left mouse button, and drag the mouse.
- To **move the object vertically up or down**, hold down the **Alt/Alt Gr** key and the left mouse button, and drag the mouse.

Note

If you are working with a *Frame* **with visible content**, the order of pressing the **Alt/Alt Gr** key and clicking the mouse button determines what *3D* does: When you hold down the **Alt/Alt Gr** key before you click the mouse, you select simulation objects located within the *Frame*. In any case, holding the **Alt/Alt Gr** key down moves the selected object vertically up or down.

If your keyboard does not have the **Alt Gr** key, hold down **Ctrl-right+Alt-right** instead.

Compare [Move an Object With Keyboard Keys](#)

Compare [Move an Object Precisely](#)

Compare [Rotate an Object Precisely](#)

Compare [Select Objects](#)

Back to [Move Objects in the Frame](#)

Move an Object With Keyboard Keys

Instead of using the mouse to manipulate the selected object, you can also hold down the **Ctrl** or **Shift** keys and press one of the arrow keys on the keyboard.

To **move** the selected object **on the grid plane**, press the designated arrow key:

- Press the **left arrow** to move the object to the left by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid Lines > Select for Snapping**.

Press the **left arrow** and hold down **Shift** to move the object to the left by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

- Press the **right arrow** to move the object to the right by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

Press the **right arrow** and hold down **Shift** to move the object to the right by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

- Press the **up arrow** to move the object upward by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

Press the **up arrow** and hold down **Shift** to move the object upward by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

- Press the **down arrow** to move the object downward by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

Press the **down arrow** and hold down **Shift** to move the object downward by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

To **move** the selected object **vertically up or down**, hold down **Ctrl** and press the designated arrow key:

- Hold down **Ctrl** and press the **up arrow** to move the object upward in the z-direction by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings**

Hold down **Shift** and **Ctrl** and press the **up arrow** to move the object upward in the z-direction by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

- Hold down **Ctrl** and press the **down arrow** to move the object downward in the z-direction by 10% of the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

Hold down **Shift** and **Ctrl** and press the **down arrow** to move the object downward in the z-direction by the distance set in the dialog **Edit Grid and Base Plate Settings** under **Grid lines > Select for Snapping**.

To **rotate** the selected object, hold down **Ctrl** and press the designated arrow key:

- Hold down **Ctrl** and press the **left arrow** to rotate the object counterclockwise by 1°.

Hold down **Shift** and **Ctrl** and press the **left arrow** to rotate the object counterclockwise by 45°.

- Hold down **Ctrl** and press the **right arrow** to rotate the object clockwise by 1°.
- Hold down **Shift** and **Ctrl** and press the **right arrow** to rotate the object clockwise by 45°.

Compare [Move an Object With the Mouse](#)

Compare [Edit Length-oriented Objects with Keyboard Keys](#)

Back to [Move Objects in the Frame](#)

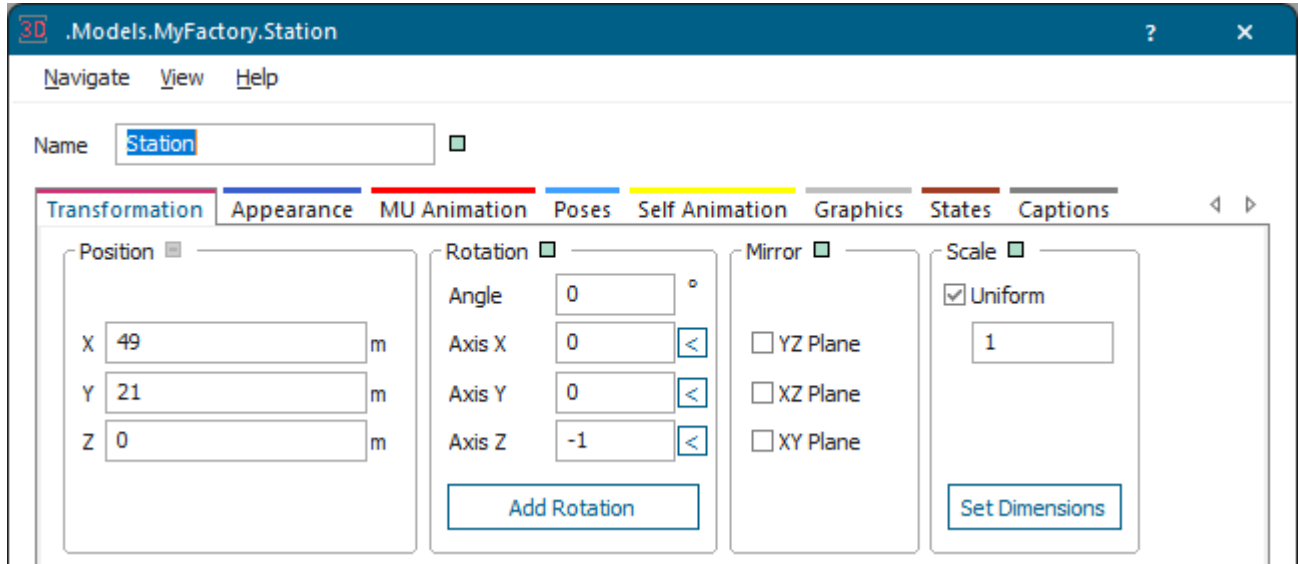
Move an Object Precisely

At times manually moving an object with the mouse or with keyboard keys will not be accurate enough for your purposes. Then, you can enter exact values into a dialog.

To do this, enter exact values into the text boxes in the dialog **3D Properties** > **Tab Transformation** to fine-tune the position, the rotation, and the scale of the selected object:

To precisely move an object:

- Select the object you would like to move.
- Press the spacebar or click  on the **Home** ribbon tab and click the tab **Transformation**.



- To move the object to the **right** on the **x-axis**, click in the text box and roll the mouse wheel forward to increase the position in steps of 0.1 meters. Hold down **Shift** and roll the mouse wheel forward to increase the position in steps of 1 meter.

To move the object to the **left** on the **x-axis**, click in the text box and roll the mouse wheel backward to decrease the position in steps of 0.1 meters. Hold down **Shift** and roll the mouse wheel forward to decrease the position in steps of 1 meter.

- Use the same procedure for the **y-axis**.
- Use the same procedure for the **z-axis**.
- Click **Apply** to apply your change and to see if they meet your requirements.
- Click **OK** once you're satisfied with the results.

Compare [Edit Grouped Graphics](#)

Back to [Move Objects in the Frame](#)

Rotate Objects in the Scene

Rotate Objects in the Frame

Plant Simulation provides several ways of rotating an object in the 3D window.

You can:

- [Rotate an Object with Keyboard Keys](#)
- [Rotate an Object Precisely](#)

Rotate an Object with Keyboard Keys

When you manually rotate an object around an axis by holding down **Shift+Ctrl** and pressing the **left** or the **right arrow** on the keyboard, *3D* uses the settings under **Rotation** in the dialog **3D Properties > Transformation > Settings for the Rotation**. You can either use the default settings or define the settings that meet your modeling needs.

To manually rotate an object in the 3D window with the settings you define:

- Click  on the **View** ribbon tab to show the grid.
- Click the object you want to rotate.
- To rotate the object by 1° to the left, hold down **Ctrl** and press the **left arrow**.
To rotate the object by 1° to the right, hold down **Ctrl** and press the **right arrow**.
- To rotate the object by 45° to the left, hold down **Ctrl** and the **Shift** and press the **left arrow**.
To rotate the object by 45° to the right, hold down **Ctrl** and **Shift** and press the **right arrow**.
- To rotate the object by 10° on the x-axis to the left, press **7** on the numeric keypad.
To rotate the object by 10° on the x-axis to the right, press **8** on the numeric keypad.
- To rotate the object by 10° on the y-axis to the left, press **4** on the numeric keypad.
To rotate the object by 10° on the y-axis to the right, press **5** on the numeric keypad.

- To rotate the object by 10° on the z-axis to the left, press **1** on the numeric keypad.
To rotate the object by 10° on the z-axis to the right, press **2** on the numeric keypad.

To define settings for manually rotating an object:

- Select **3D Properties** > **Transformation** > **Settings for the Rotation** on the **Home** ribbon tab.

- Define the settings how 3D rotates the object in the text boxes next to **Axis**.
You can either click one of the buttons to rotate the object around that axis, or you can set the axis yourself.
- To rotate the object around an arbitrary axis, defined by the x-component, y-component and z-component of this axis:
Enter a value for the x-component of the rotation axis.
Enter a value for the y-component of the rotation axis.
Enter a value for the z-component of the rotation axis.
- Then, click **Apply** to apply these settings, so that they are active when you rotate the object.

Note

Transformations of an object, such as moving, rotating or scaling it, in one 3D window **are** not immediately **updated** in all other 3D windows, but **only after you deselect the changed object** in its 3D window.

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Rotate an Object Precisely

At times manually rotating an object with keyboard keys will not be accurate enough. Then, you can enter exact values into the text boxes in the dialog **3D Properties** > **Tab Transformation** to fine tune the position, the rotation, and the scale of the selected object.

To precisely rotate an object on the **Tab Transformation**:

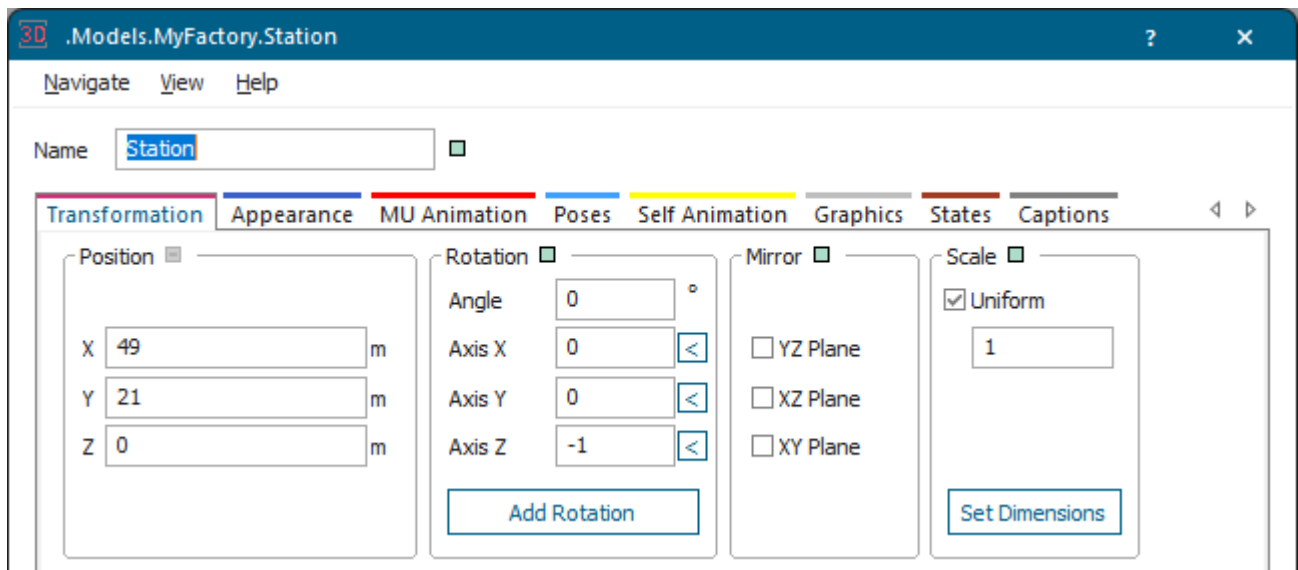
- Click it with the left mouse button to select it.

- Press the spacebar or click  on the **Home** ribbon tab.
- To rotate the object around an arbitrary axis, defined by its x-component, y-component and z-component:

Enter a value for the x-component of the rotation axis.

Enter a value for the y-component of the rotation axis.

Enter a value for the z-component of the rotation axis.



- Type in the offset to the center of the object for the rotation around the x-axis, the y-axis, and the z-axis. Normally you do not type in another value but apply the presets of 0, 0, 0. This rotates the object around its center.
- Enter the **angle of rotation** in degrees into the text box.
- To apply your changes if you are satisfied with the results of your changes, click **Apply**.

Transformations of an object, such as moving, rotating or scaling it, in one 3D window **are** not immediately **updated** in all other 3D windows, but **only after you deselect the changed object** in its 3D window.

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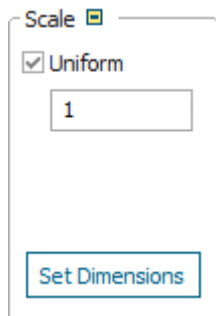
Scale Objects in the Frame

You can scale objects in the *Frame* containing your simulation model.

To scale an object in the 3D window, type exact values into the text boxes in the dialog **3D Properties** > **Tab Transformation** > **Scale** to fine tune the position, the rotation, and the scale of the selected object:

- Click it with the left mouse button to select it.

Press the spacebar or click  on the **Home** ribbon tab.

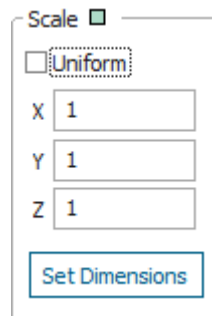


Scale 

☒ Uniform

1

Set Dimensions



Scale 

☐ Uniform

X 1

Y 1

Z 1

Set Dimensions

- Select **Uniform** to scale the object with the same factor for all three dimensions and to not distort it while scaling it. This automatically enters the value you type into one text box into the remaining two text boxes. This way you do not have to manually type the same value into each text box again and again.

Type in a value into the text boxes for the x-axis, the y-axis and the z-axis to scale the object on the respective axis with that factor. Typing 0.5, for example, halves the size of the object on the axis for which you typed it, typing in 2 doubles its size on that axis.

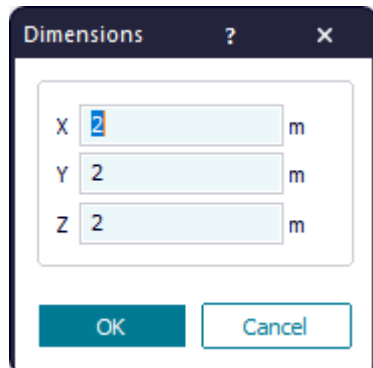
Note

You can only type in non-zero values.

- Clear **Uniform** to scale the object with the factors you type in for the different dimensions. When you type in different values, it does distort the object, for example lengthens or shortens it.

Type a value into the text boxes for the x-axis, the y-axis and the z-axis to scale the object with that factor on the respective axis.

- Click **Set Dimensions**. Then type in the physical size of the scaled object considering the graphics of the object.



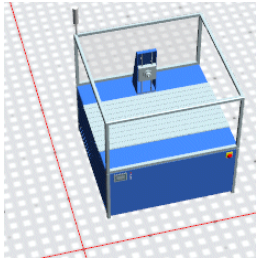
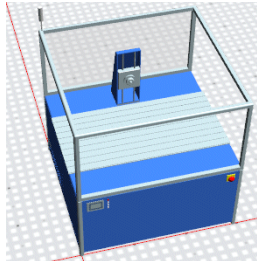
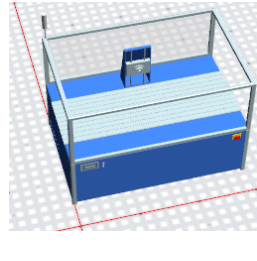
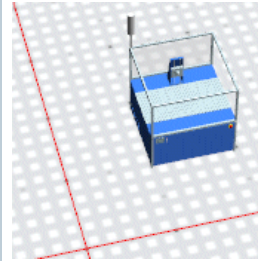
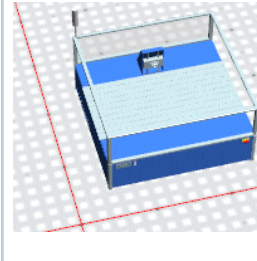
Dimensions ? x

X 2 m

Y 2 m

Z 2 m

OK Cancel

Not scaled	Scale Uniform 1.5	Scale Not Uniform, X=1.5	Set Dimensions Uniform 0.5	Set Dimensions Not Uniform, X/ Y=1, Z=0.5
				

Transformations of an object, such as moving, rotating or scaling it, in one 3D window **are** not immediately **updated** in all other 3D windows, but **only after you deselect the changed object** in its 3D window.

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Work with Length-oriented Objects

Work with Length-oriented Objects

The length-oriented objects use their length and their dimensions during the simulation.

Plant Simulation provides these length-oriented objects: The [Conveyor](#), the [Track](#), the [TwoLaneTrack](#), the [Footpath](#), the [Container](#), the [Transporter](#), and the [Pipe](#).

Plant Simulation generates the graphics automatically according to their settings for these objects and for the [Turnplate](#), the [Turntable](#), the [Converter](#), the [AngularConverter](#), and the [Store](#).

To create a class from a *Conveyor*, *Turntable*, *Track*, *TwoLaneTrack*, or *FootPath*, model this class in the *Frame*, and then drag this object from the *Frame* to the *Class Library*. You can then insert this class into the active and into any other simulation model.

You can:

- [Insert Curved and Straight Segments](#)
- [Auto-connect Length-oriented Objects](#)
- [Draw Straight and Curved Segments with a 90° Angle](#)
- [Draw Straight and Curved Segments without Fixed Values](#)
- [Change the Shape of a Segment](#)
- [Create a Curved Object with SimTalk Commands](#)

- [Reuse the Segments of a Length-oriented Object](#)
- [Keyboard Shortcuts for Inserting Straight and Curved Segments of an Object](#)
- [Edit Length-oriented Objects with the Mouse](#)
- [Edit Length-oriented Objects with Keyboard Keys](#)
- [Edit Length-oriented Objects Precisely](#)
- [Model Differences in Height Between Conveying Elements](#)
- [Model a Spiral Conveyor](#)

Compare [Generated and Non-Generated Graphics](#)

Compare [Change the Shape of a Segment](#)

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Insert Curved and Straight Segments

You can insert a *Track*, a *TwoLaneTrack*, a *Conveyor*, a *FootPath*, or a *Pipe* as a sequence of curved and straight segments into the *Frame* that contains your simulation model.

Proceed as follows:

- Zoom the *Frame* to a size that it can accommodate the object you want to insert.
- Click the *length-oriented object* in the *Toolbox*, and drag the mouse to the *Frame* window and click the point where the object is to start. This opens the dialog **Edit Parameters of Curve**.

To draw a straight segment, click a second time at the position at which you want the end of the segment to be. To draw a curved segment, hold down **Ctrl**, drag the mouse down to draw the curve downward, and click the left mouse button once to set the curve.

Line/Arc Parameters

Line Segments

Tangential Angle

0.000 °

☐ fixed

Line Length

9.400 m

☐ fixed

Arc Segments

Arc Length

3.142 m

☐ fixed

Curve Angle

90 °

☒ fixed

Radius

2 m

☒ fixed

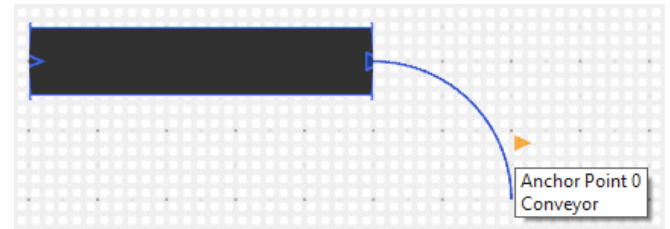
Anchor Point Height

0 m

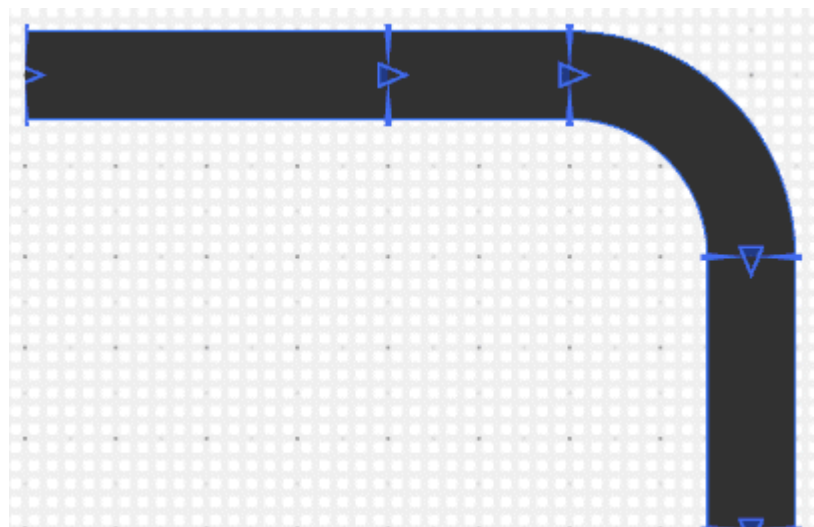
Finish

Abort

Delete Last Point



To draw the next straight segment of the line, release **Ctrl**, drag the mouse to the end point of the line and click the left mouse button once.



To draw a continuous sequence of straight and curved segments, for example for continuous conveyors, click an existing curved or straight end segment to start a new segment. *Plant Simulation* then uses the end coordinate of the first sequence as the start coordinate of the second sequence and automatically connects the two. You can prevent this by holding down **Alt**.

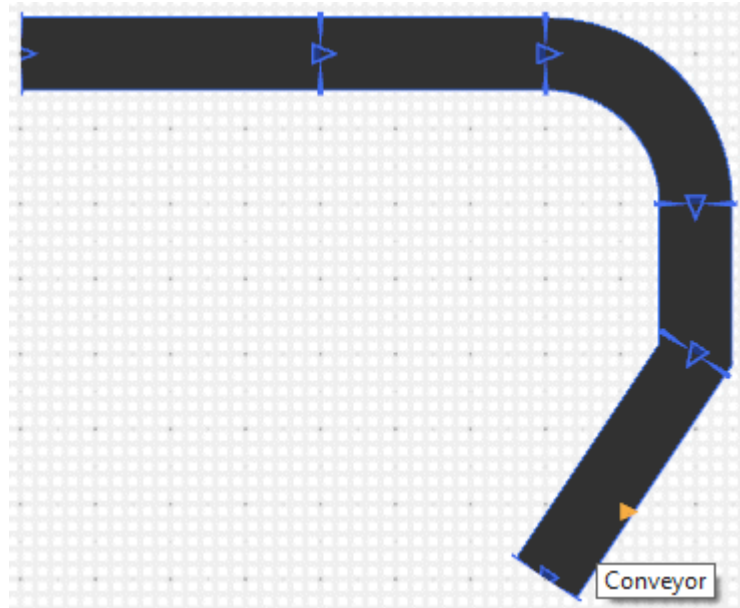
Click the right mouse button once or click **Finish** Finish to exit **Insert** mode and to insert the curve you drew. This also closes the dialog **Edit Parameters of Curve**.

Press **Esc** or click **Abort** Abort to terminate curve mode without inserting the curve.

Click **Delete Last Point** Delete Last Point to delete the most recently inserted anchor point.

When the grid is active , *Plant Simulation* places the points that make up the line onto grid points, provided this is possible. For this, fixed values take precedence. When you type in a radius of 2.5 meters, for example, the end point of the arc will not be placed onto a grid point. This results in a non-tangential transition between the end of the arc and the end of the next straight segment, which will be placed onto a grid point again.

When you insert a straight segment after a curved segment, *Plant Simulation* always enforces a tangential angle of 0°.

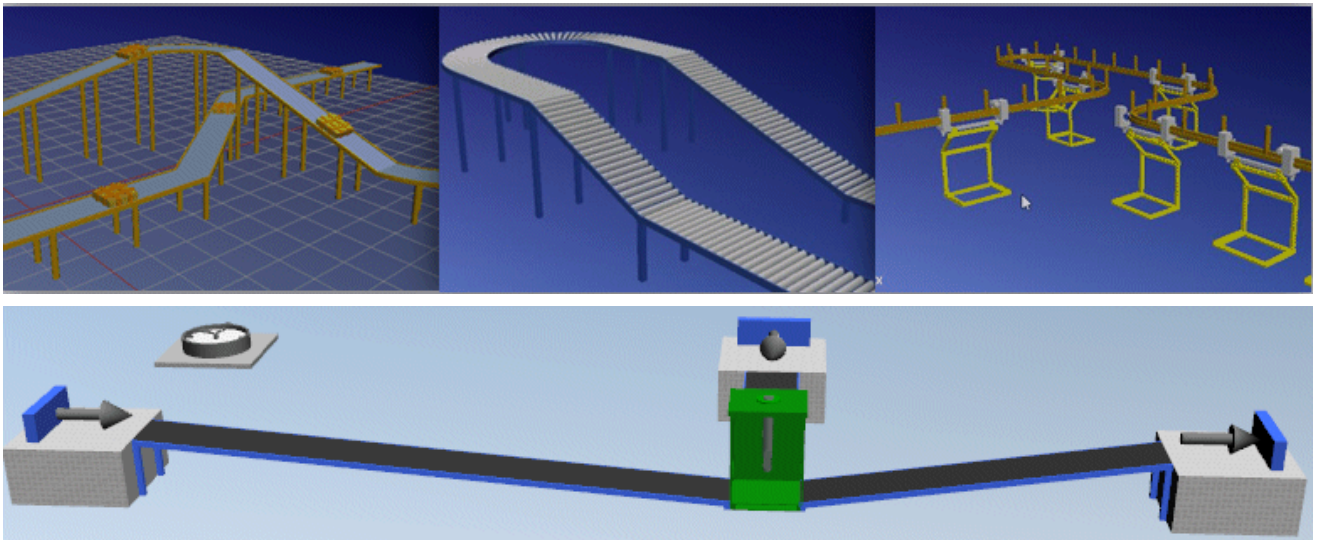


To insert the *length-oriented curved object* with fixed values, type them into the text boxes in the dialog and select the check box **fixed** of the respective item to tell *Plant Simulation* that you want to use these fixed values.

- For a line segment, you can type in the **Line Length** in world coordinates and the **Tangential Angle** as compared to the previous segment.

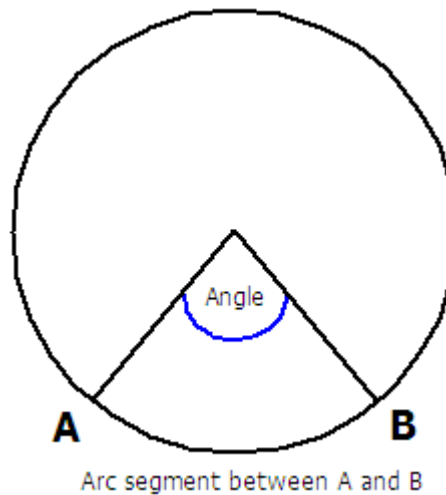
Line/Arc Parameters			
Line Segments			
Tangential Angle	0.000	°	☐ fixed
Line Length	9.400	m	☐ fixed
Arc Segments			
Arc Length	3.142	m	☐ fixed
Curve Angle	90	°	☒ fixed
Radius	2	m	☒ fixed
Anchor Point Height	0	m	
Finish		Abort Delete Last Point	

For both segment types you can type in an **Anchor Point Height**. It determines the height, i.e., the distance from the floor, to the next anchor point. It is, for example, useful when you model a ramp for material handling equipment.



- For a **curved segment**, you can type in the **Arc Length**, its **Radius**, and its **Curve Angle** in degrees as compared to the previous segment.

The **Curve Angle** is the angle that is formed by the radius between the start point A and the end point B of the arc segment. A positive number designates a clockwise angle. A negative number a counterclockwise angle.

**Note**

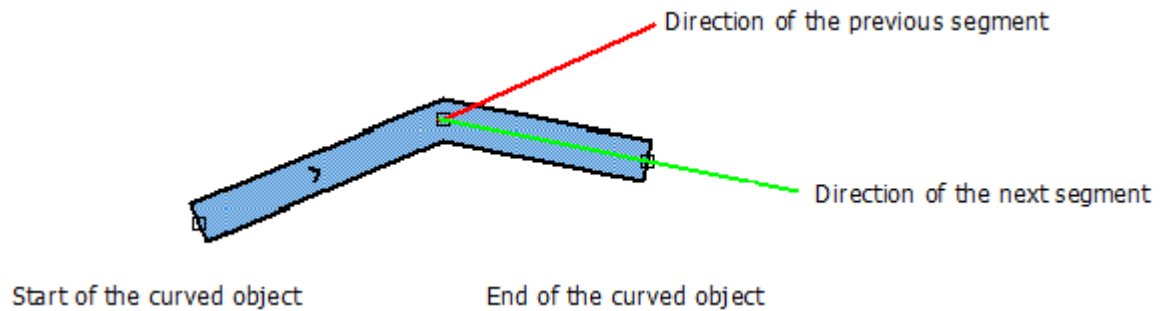
When you define a curve angle of 360°, *Plant Simulation* automatically connects the end point the curve with its starting point.

- For a following **Straight Segment** you can type in the **Tangential Angle** and the **Line Length**.

Line/Arc Parameters			
Line Segments			
Tangential Angle	0.000	°	☐ fixed
Line Length	0.000	m	☐ fixed
Arc Segments			
Arc Length	3.142	m	☐ fixed
Curve Angle	90	°	☒ fixed
Radius	2	m	☒ fixed
Anchor Point Height	1	m	
Finish Abort Delete Last Point			

The tangential angle designates the change in the tangential direction. Type in a **positive number** for a **clockwise angle**. Type in a **negative number** for a **counterclockwise angle**. Normally the tangential angle is relative to the previous direction of the curved segment. For the first segment the tangential angle is the angle to the positive x direction.

When you insert a straight segment after a curved segment, *Plant Simulation* always enforces a tangential angle of 0 degrees.



Plant Simulation creates the **start tangential angle** as follows:

After clicking an existing object with the mouse, *Plant Simulation* applies the angle from the previous object as **start tangential angle**. This also applies to icons.

When setting a straight segment by a mouse click on an existing object, *Plant Simulation* uses, depending on the mouse position, either the angle of the object or ± 90 degrees to this angle as the **start tangential angle**.

When setting a curved segment without clicking an existing object, *Plant Simulation* proceeds like this: When you type in an angle for straight segments, it uses this angle as the **start tangential angle**. When you hold down **Ctrl**, *Plant Simulation* uses, depending on the mouse position, 0, ± 90 or 180 degrees as the start tangential angle.

When you insert a length-oriented curved object, *Plant Simulation* shows the extrusion path of this object. The path of the polycurve initially just consists of the starting point:

- Either the insertion point, i.e., the position on the grid on which you click or
- The end point of the picked object, which is its exit projected onto the XY layer.
- *Plant Simulation* shows a movable line, which connects a manipulator at the starting point with another manipulator attached to the cursor.
- You insert straight path segments by clicking the left mouse button. You insert curved path segments by holding down **Shift** and by clicking the left mouse button.
- You terminate curve mode by clicking the right mouse button or by pressing **Esc**.
- *Plant Simulation* will not insert a length-oriented object, if the curved object does not consist of at least two segments/points after you finished or terminated inserting the curved object.

Compare [Add Animations and Interactions to the Simulation Objects](#)

Compare [Edit a Curved Path with the Mouse](#)

Back to [Work with Length-oriented Objects](#)

Auto-connect Length-oriented Objects

When you define a length-oriented object and click another object with the mouse to set that object as the successor of the length-oriented object, *Plant Simulation* attempts to **automatically bridge the gap** from the current end of the length-oriented object to the successor with additional straight and/or curved segments.

How Plant Simulation Bridges a Gap

To bridge the gap, *Plant Simulation* tries to create additional line and/or curve segments so that the resulting length-oriented object connects its predecessor object and its successor object tangentially.

The predecessor and the successor can be a length-oriented object, a point-oriented object, or no object at all. The direction to be continued tangentially is determined by the last/first segment of a length-oriented predecessor/successor, the **X-Axis** of a point-oriented successor, or the **X-Axis** of the scene.

Plant Simulation automatically corrects any existing difference in height in the last segment of the constructed length-oriented object.

Plant Simulation can only close a gap under the following conditions:

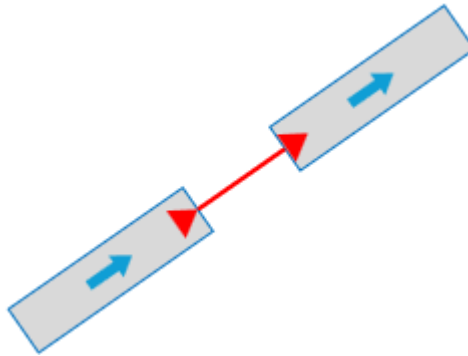
- The **Curve Angle** in the **Line/Arc Parameters** dialog is fixed to 90 degrees and the **Radius** is fixed as well.

Line/Arc Parameters			
Line Segments			
Tangential Angle	10.491 °	☐	fixed
Line Length	7.616 m	☐	fixed
Arc Segments			
Arc Length	3.142 m	☐	fixed
Curve Angle	90 °	☒	fixed
Radius	2 m	☒	fixed
Anchor Point Height	1 m		
Finish Abort Delete Last Point			

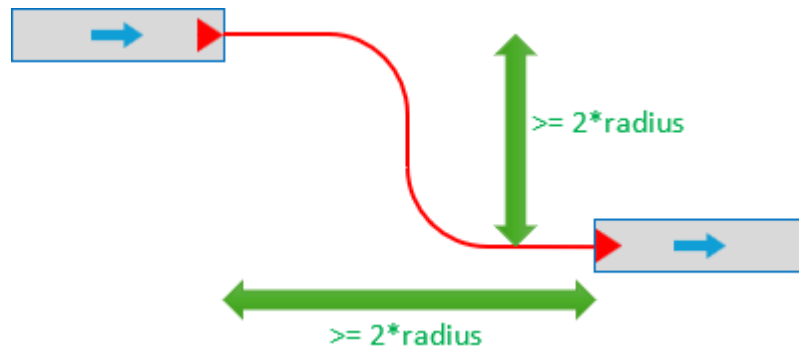
- The objects, which you want to connect, are aligned in parallel or perpendicular to each other.
- The gap can be closed with maximally two right-angled curve segments.
- The objects, which you want to connect, are not rotated around any other axis than the z-axis.

Meeting the above conditions, *Plant Simulation* can bridge a gap in the following cases. The objects do not necessarily have to be aligned to the grid:

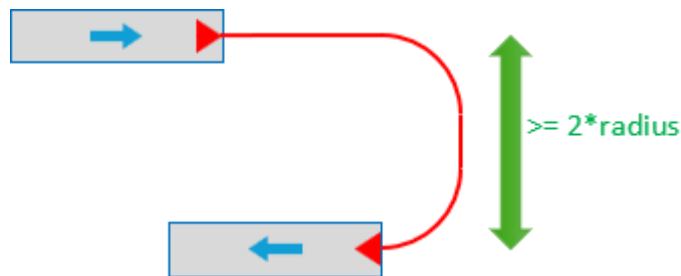
- The objects, which you want to connect, are aligned to each other and point in the same direction.



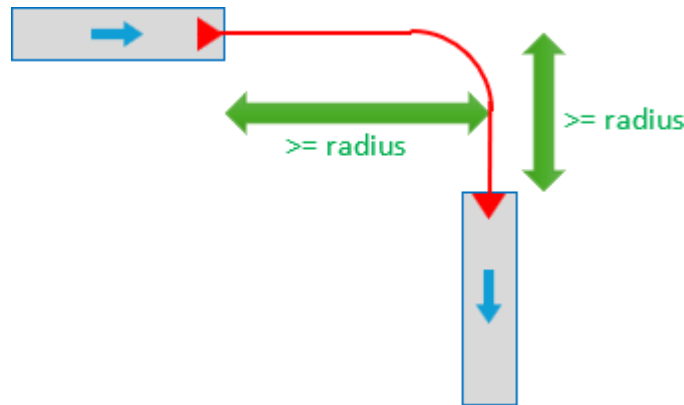
- The objects, which you want to connect, are positioned parallel to each other and point in the same direction. Both the tangential and vertical distance of the connection points is at least the double radius of the **Line/Arc Parameters**.



- The objects, which you want to connect, are positioned parallel to each other but point in the opposite direction. The vertical distance of the connection points is at least the double radius of the **Line/Arc Parameters**.



- The objects, which you want to connect, are placed with a right angle to each other. Both the tangential and vertical distance of the connection points is at least the radius of the **Line/Arc Parameters**.



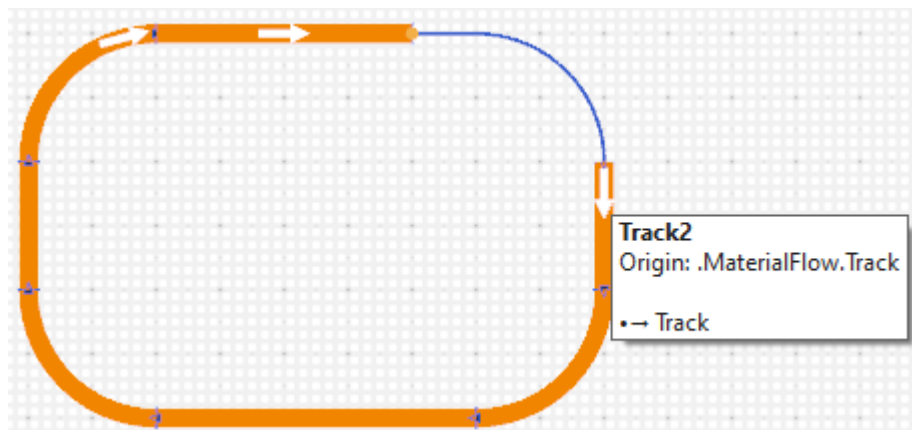
If *Plant Simulation* cannot close the gap under these conditions, it suggests a straight line pointing directly to the connection point of the successor object, provided you did not hold the **Ctrl** key down.

You can then decide what you want to do:

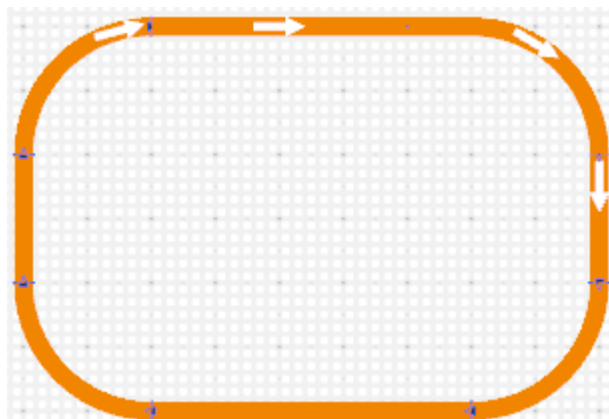
- Connect the length-oriented object with segments along the suggested polyline by clicking the successor object.
- Move the cursor away from the successor object and manually define separate tangentially prolongating segments, compare [Insert Curved and Straight Segments](#).

How to Bridge and Close the Gap

If *Plant Simulation* can bridge and close the gap with the picked successor object, then it shows the computed polyline as a preview as soon as you drag the cursor over the successor object.



Click the successor, to close the gap and establish the connections.



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Draw Straight and Curved Segments with a 90° Angle

You can draw a *length-oriented curved object* with a straight segment, and a 90° curve followed by another straight or curved segment.

Proceed as follows:

- Click the icon  if the *Conveyor* in the *Toolbox*.
- Drag the mouse to the *Frame* window to the point where you would like to start the straight segment of the line.
- Click the left mouse button once to set the starting point of the line.

Line/Arc Parameters

Line segments

Tangential angle: ° ☐ fixed

Line length: m ☐ fixed

Arc segments

Arc length: m ☐ fixed

Curve angle: ° ☒ fixed

Radius: m ☒ fixed

Anchor point height: m

Finish

Abort

Delete Last Point



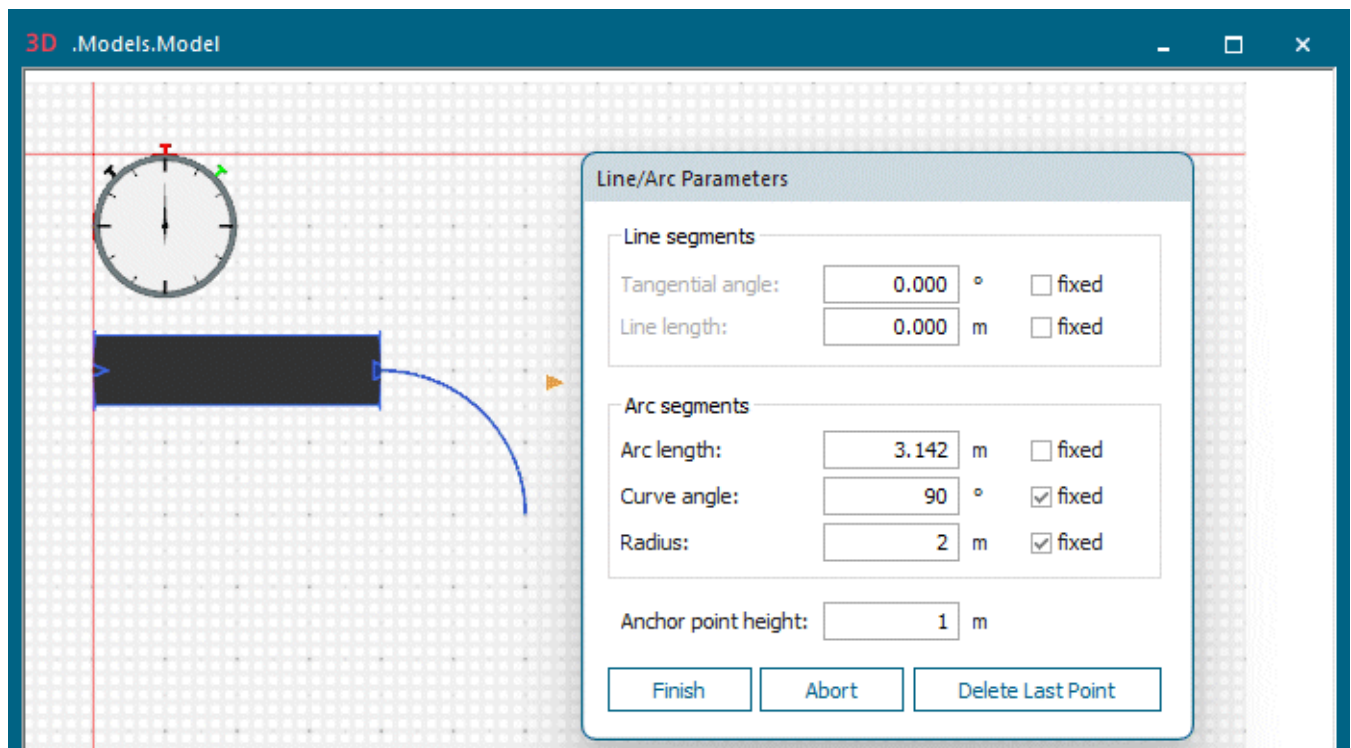
- *Plant Simulation* opens the dialog **Edit Parameters of Curve**. Here, we defined default settings for inserting a curve with a 90° angle, i.e., a fixed tangential angle of 0°, a fixed curve angle of 90° and a fixed radius of 2 meters.

You can either apply these values or you can type in a different radius to make the curve wider. To do so, drag the mouse to the dialog and type another value into the text box.

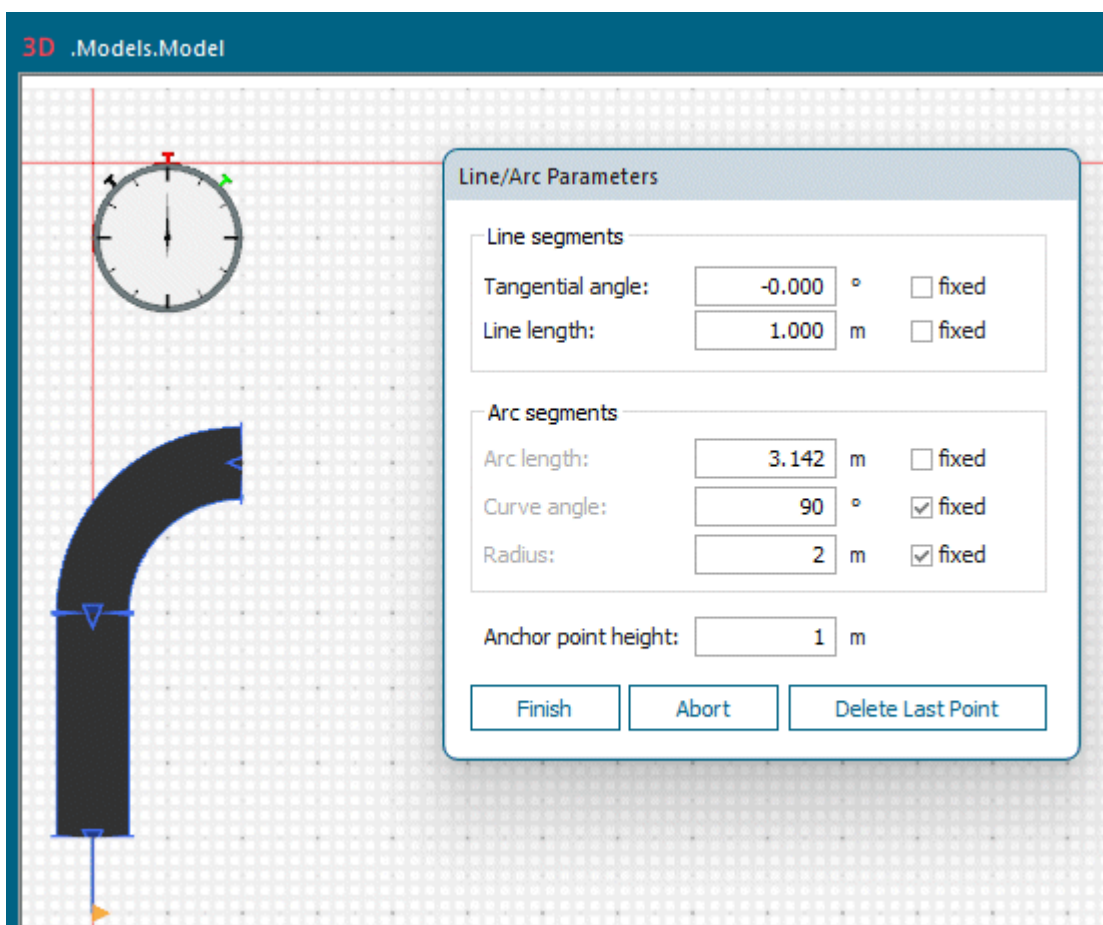
Note

Plant Simulation saves the last set of settings in the dialog, so that you can re-use them the next time you insert a curved object.

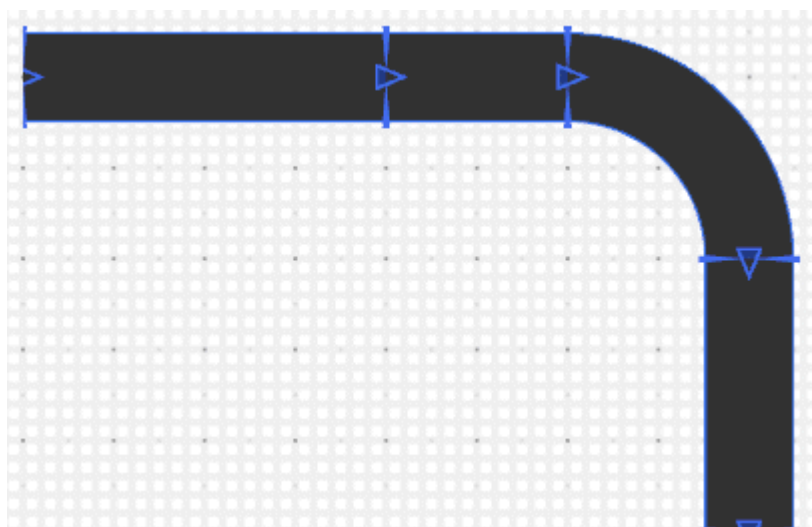
To draw the curve, drag the mouse back to the *Frame* window, hold down **Ctrl**, drag the mouse down to draw the curve downward, and click the left mouse button once to set the curve.



To draw a curve pointing to the left: Drag the mouse back to the *Frame* window and hold down **Ctrl**. Make sure that **Fixed** for the **Curve angle** of the arc segments is activated and type in 90. Then drag the mouse down to draw the curve downward, and click the left mouse button once to set the curve.

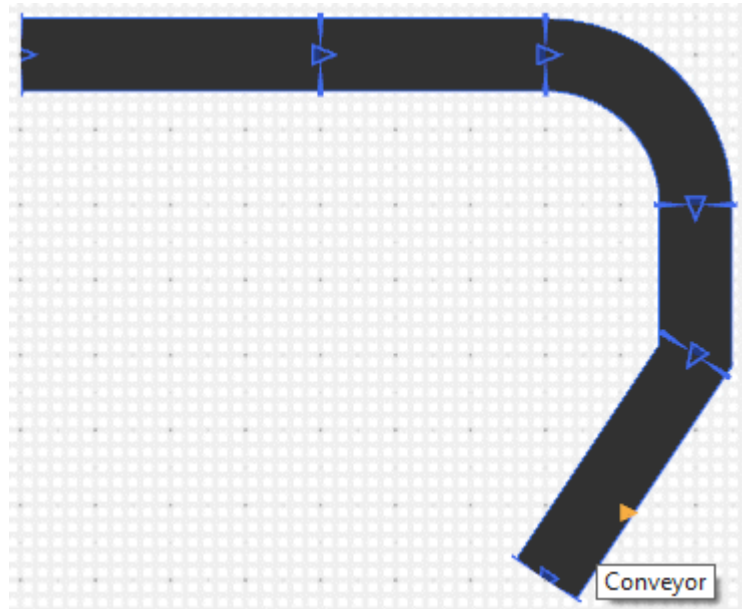


- To draw the next straight segment of the line, release **Ctrl**, drag the mouse to the end point of the line and click the left mouse button once.



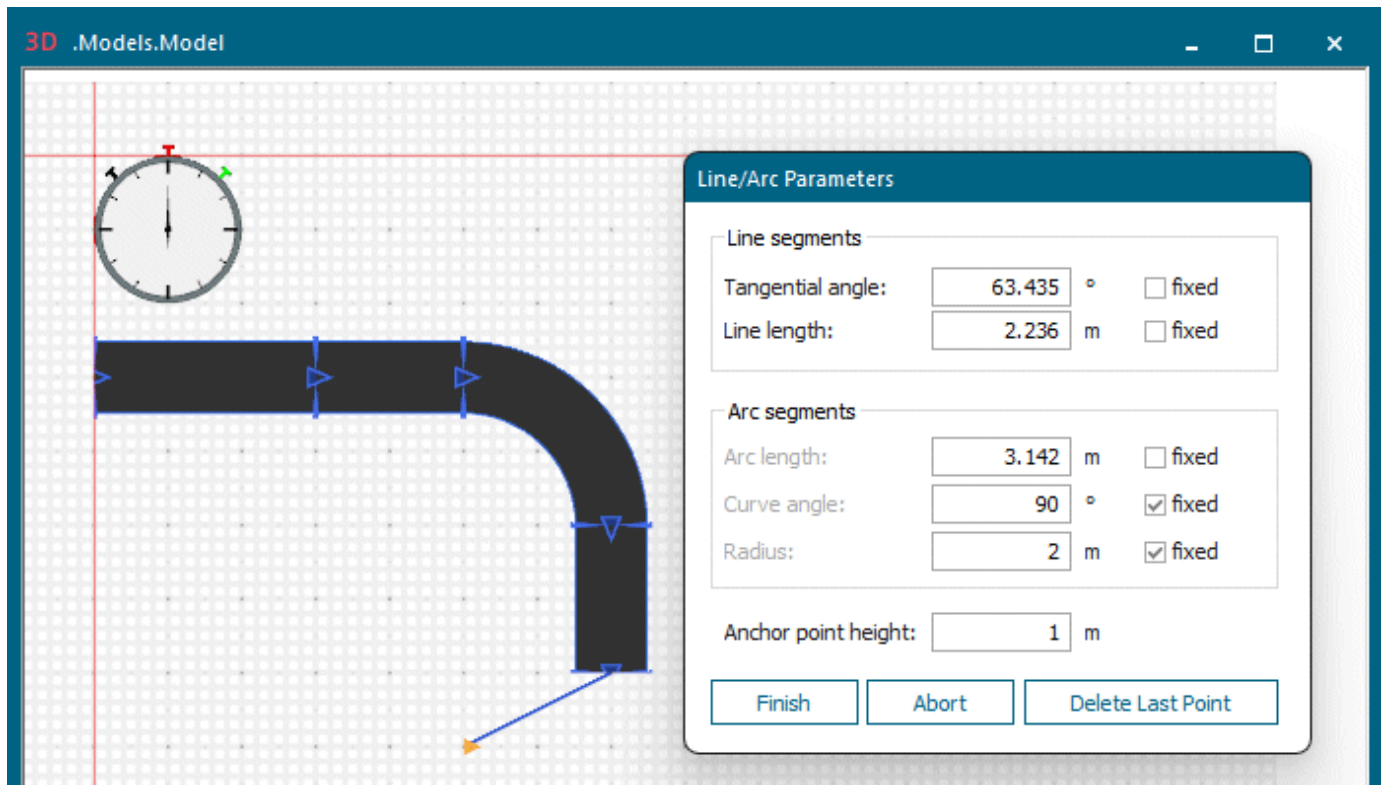
- Click the right mouse button once to exit **Insert** mode. This also closes the dialog **Edit Parameters of Curve**.

When the grid is active , *Plant Simulation* places the points that make up the line onto grid points, provided this is possible. For this fixed vales take precedence. When you type in a radius of 2.5 meters, for example, the end point of the arc will not be placed onto a grid point. This results in a non-tangential transition between the end of the arc and the end of the next straight segment, which will be placed onto a grid point again.



To get a tangential transition, you can:

- Deactivate **Snap to Grid** by holding down **Alt**, when you click the left mouse button to set the end point of the second straight segment.
- Type in a **Tangential angle** of 0 degrees and set it as **fixed** for this straight segment. Then deactivate this setting for the following straight segments.



To draw a curved segment without using fixed values, you have to click the left mouse button three times. To activate curve mode, hold down **Ctrl**, when you click the mouse button the first time!

- The first mouse click sets the starting point.
- The second mouse click sets the radius.
- The third mouse click sets the arc length or the curve angle respectively.

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Draw Straight and Curved Segments without Fixed Values

You can draw a straight and a curved segment of a *length-oriented curved object* without using fixed values.

Proceed as follows:

- To set the starting point of the segment, click the left mouse button once. You will notice that *Plant Simulation* attaches the icon of the *Connector* to the front end of the cursor  in the *Frame* window. The text boxes now show the actual values of the line segment.
- Drag the mouse in a direction of your choice and as far as you want the segment to be long. Click the left mouse button again to set the first anchor point. Now, *Plant Simulation* draws the segment with the settings you selected.

- Continue this until the object has the length and the shape you want.
- To set the end point of the segment, and to exit **Insert** mode, click the right mouse button.

To draw a curved segment of a *length-oriented curved object* without using fixed values:

- To set the starting point of the curved segment, hold down **Ctrl** and click the left mouse button once.
You will notice that the text boxes now show the actual values of the curve segment.
- To set the **Tangential Angle**, drag the mouse in a direction of your choice.
- To set the **Radius** of the curved segment continue dragging the mouse until the curve has the radius you want it to be and click the left mouse button.
- To set the **Arc Length**, continue dragging the mouse until the curve is as long as you want it to be and click the left mouse button.
- To set the end point of the segment, click the right mouse button.
- To delete the last anchor point from the curve, drag the mouse pointer from the *Frame* window to the dialog **Edit Parameters of Curve** and click **Delete Last Point**.

Naturally you can combine both techniques to insert curved and straight segments in any sequence you like.

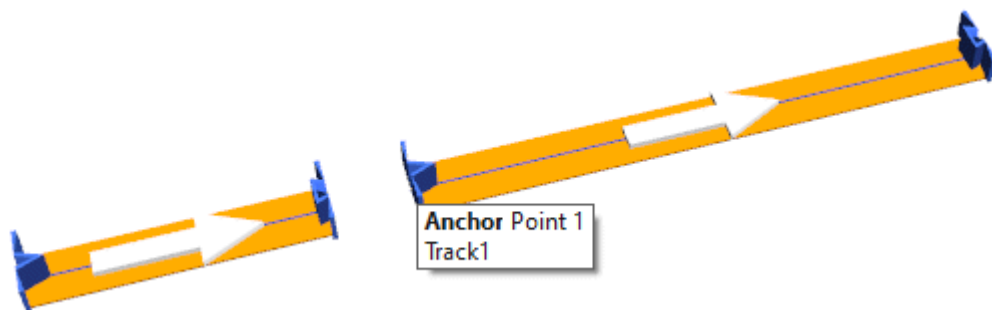
Back to [Work with Length-oriented Objects](#)

Change the Shape of a Segment

You can modify the shape of a segment in a number of ways.

Proceed as follows:

- You can change the shape of a straight segment by clicking [Show Manipulators](#) and by then dragging the manipulators.

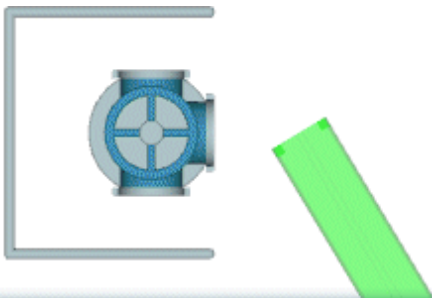


- To delete an anchor point from the curve, right-click it and select [Delete Anchor Point](#) on the context menu.
- To extend an existing curved segment of the *length-oriented object* without adding a new anchor point, click the drag point with the left mouse button and drag the mouse to the new position.

- To append anchor points to the end of the curve, right-click anywhere on the curve and select **Segments > Append [segment]** on the context menu. Add an additional anchor point with the dialog **Edit Parameters of Curve**.
- To edit existing segments and to insert new segments, select **Segments > Edit [segment]**. In the **Segments Table** you can also insert a segment at the beginning of the object which is not possible manually with the mouse.

In the example below we insert a curved segment at the beginning of the *Pipe* so that it seamlessly attaches to the *FluidSource*.

- Click the *Pipe* and press **F7**.



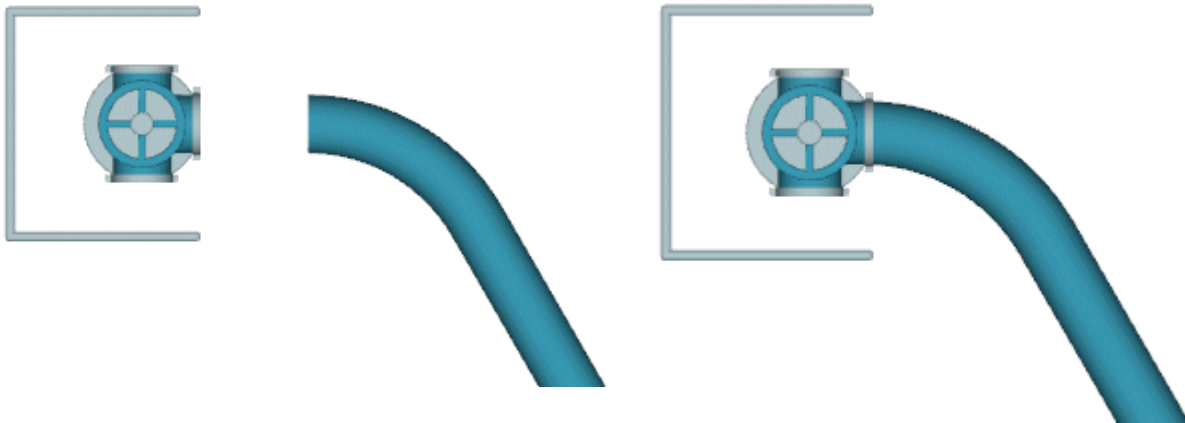
I..	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	V.	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	11.400	12.000	0.500	0.000
2	59.0362434679265	5.830951894...				0	14.400	7.000	0.500	5.831

- This opens the **Segments** table. Click in row 2 and select **Insert Row**. This inserts an empty row above row 2.

We typed in these values:

I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	10.400	10.000	0.500	0.000
2	0		60	1	☐	0	11.266	9.500	0.500	1.047
3	0	5.2				0	13.866	4.997	0.500	6.247

- The result looks like this. We then have to adjust the position of the *Pipe*.



- To **change the direction** of a curved object, for example from right to left, edit the **Curve angle** in the **Segments** table. Our *Conveyor1* evades an obstacle to the right.



The **Curve angles** in the **Segments** table look like this.

I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	8.000	-12.000	1.000	0.000
2	0	3				0	11.000	-12.000	1.000	3.000
3	0		90	2	☐	0	13.000	-14.000	1.000	6.142
4	0		-90	2	☐	0	15.000	-16.000	1.000	9.283
5	0		-90	2	☐	0	17.000	-14.000	1.000	12.425
6	0		90	2	☐	0	19.000	-12.000	1.000	15.566
7	0	2				0	21.000	-12.000	1.000	17.566

OK Cancel Apply

Our *Conveyor2* is to evade the obstacle to the left. To accomplish this, we changed the **Curve angles** in the **Segments** table as shown below.

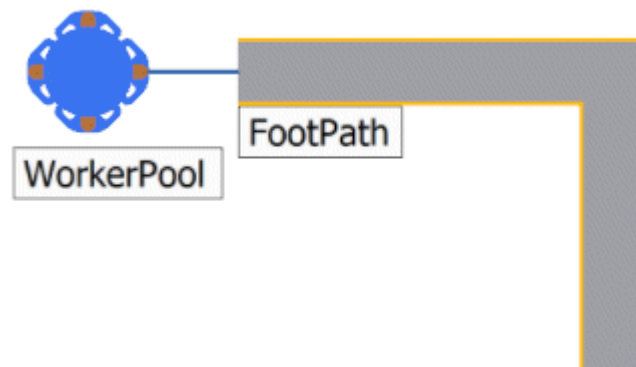
I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	8.000	-25.000	1.000	0.000
2	0	3				0	11.000	-25.000	1.000	3.000
3	0		-90	2	☐	0	13.000	-23.000	1.000	6.142
4	0		90	2	☐	0	15.000	-21.000	1.000	9.283
5	0		90	2	☐	0	17.000	-23.000	1.000	12.425
6	0		-90	2	☐	0	19.000	-25.000	1.000	15.566
7	0	2				0	21.000	-25.000	1.000	17.566

OK Cancel Apply

This then looks like this:



- To replace the right angle of a curved object with a curve, we are going to edit the **Segments** table. We originally inserted this *Footpath* with a right angle. We now want the *Worker* to walk along a curve.



We press **F7** to open the **Segments** table of the *Footpath*. It looks like this:

Segments – .Models.CornerToCurve.FootPath

I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	-10.000	4.000	0.000	0.000
2	0	5				0	-5.000	4.000	0.000	5.000
3	90	4				0	-5.000	0.000	0.000	9.000

OK Cancel Apply

To see how we have to change the **Segments** table of our *Footpath*, we insert a second *Footpath* with a curve instead of the right angle above our *Footpath*. *Footpath1* and its **Segments** table look like this:

Segments – .Models.CornerToCurve.FootPath1

I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	-10.000	4.000	0.000	0.000
2	0	3				0	-7.000	4.000	0.000	3.000
3	0		90	2	☐	0	-5.000	2.000	0.000	6.142
4	0	2				0	-5.000	0.000	0.000	8.142

OK Cancel Apply

Then left-click the *Footpath* and press **F7**. Right-click in the last row of the **Segments** table and select **Insert Row** to insert an empty row before it. Type in the values that our curved *Footpath1* has. Click **Apply**. Our *Footpath* now has a curve instead of a right-angle.

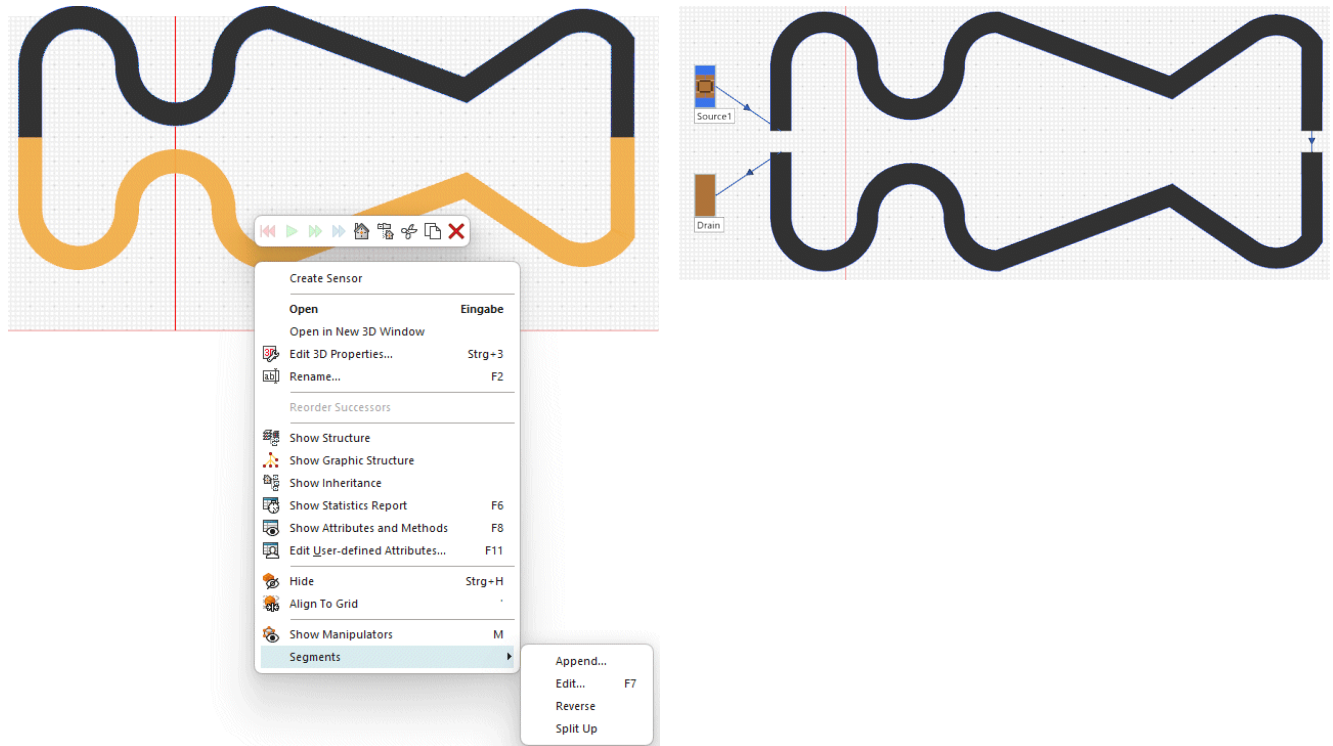
Segments – .Models.CornerToCurve.FootPath

I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	-10.000	4.000	0.000	0.000
2	0	3				0	-7.000	4.000	0.000	3.000
3	0		90	2	☐	0	-5.000	2.000	0.000	6.142
4	0	2				0	-5.000	0.000	0.000	8.142

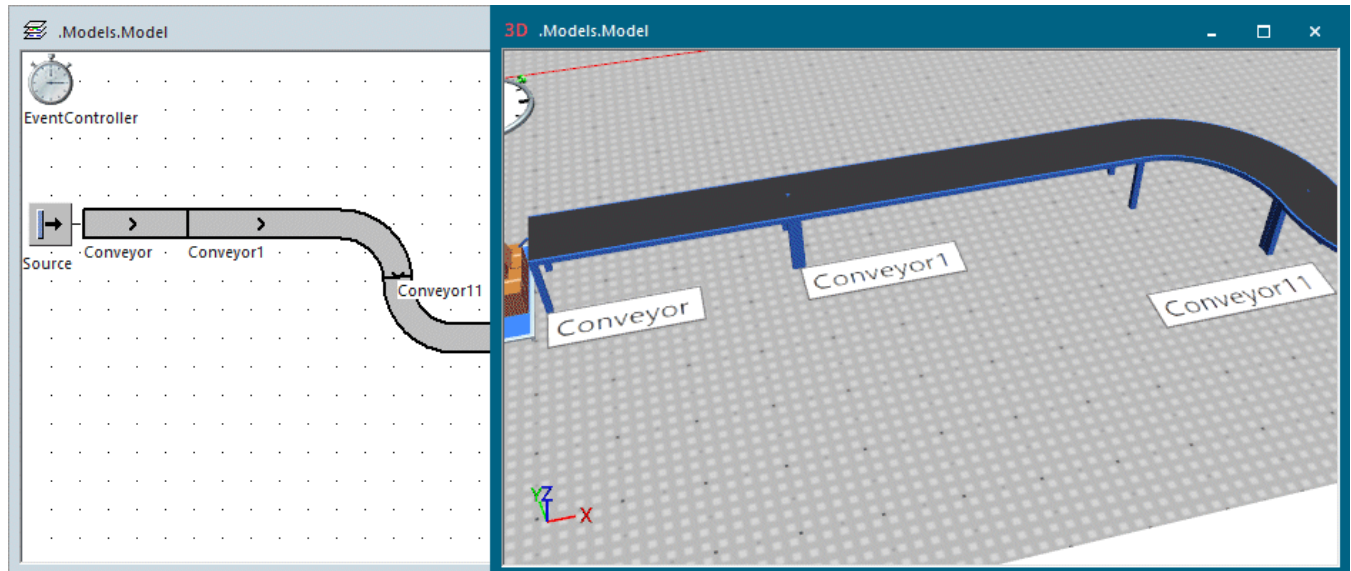
OK Cancel Apply

You can then delete *Footpath1* which just provided the template for our curve.

- To change the direction of motion of a curved object, i.e., to make the last point of the object the first point, right-click the object and select **Reverse [segment]**. This is especially handy if you want to mirror the layout of a production line by copying it and then reversing the direction of motion of the mirrored line to ensure a circular flow of materials.



- To split the *length-oriented curved objects* into individual separate segments at the selected point, click the point where you would like to split it up and select **Split Up [segment]**. This looks like this:



- To move the entire *length-oriented curved object*, click it once with the left mouse button and drag the mouse up or down or left or right. You can also use the arrow keys to move the object one pixel at a time or hold down **Shift** and press an arrow key to move the object one grid unit at a time.
- To delete the entire *length-oriented curved object*, click it once with the left mouse button and press **Delete**.

- To extend an existing *length-oriented curved object*, without adding a new anchor point, grab the handle  with the left mouse button and drag the mouse to the new location.
- To link two or more existing *length-oriented curved object*, insert a *Connector* between them.
- To insert and connect another straight length-oriented object with an existing *length-oriented curved object*, click with the left mouse button on the end of the first object, move the mouse to the beginning of the second object and click the mouse button.

Compare [Work with Length-oriented Objects](#)

Compare [Change the Shape of the Pipes](#)

Back to [Work with Length-oriented Objects](#)

Create a Curved Object with SimTalk Commands

You can also create a curved object with the *SimTalk* commands.

To do so, use the methods [derive](#) and [duplicate](#).

```
var obj := .Materialflow.Conveyor.derive(.Models.Frame)
obj.Coordinate3D := [5, 5, 0]

var obj1 := .Materialflow.Conveyor.derive(.Models.Frame)
obj1.Coordinate3D := [1, 1, 0]
```

[Reuse the Segments of a Length-oriented Object](#) shows how to use the methods [getCurveSegments \[SimTalk\]](#) and [setCurveSegments \[SimTalk\]](#).

Back to [Work with Length-oriented Objects](#)

Reuse the Segments of a Length-oriented Object

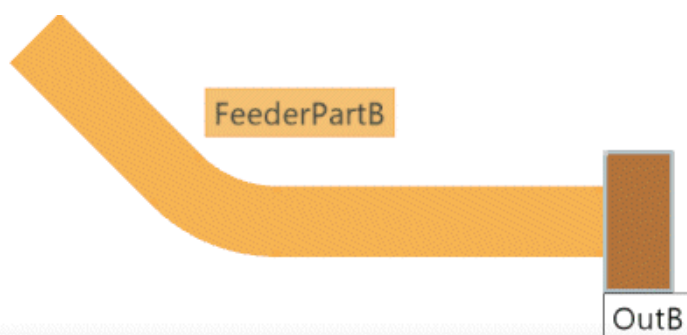
For smaller segments tables you can open two simulation models at the same time, copy the values in one model and paste them into the segments table of the other model. This way you do not have to type in the values a second time.

In the example below we copy the values of a feeder line for a *Drain* from a German model to an English model.



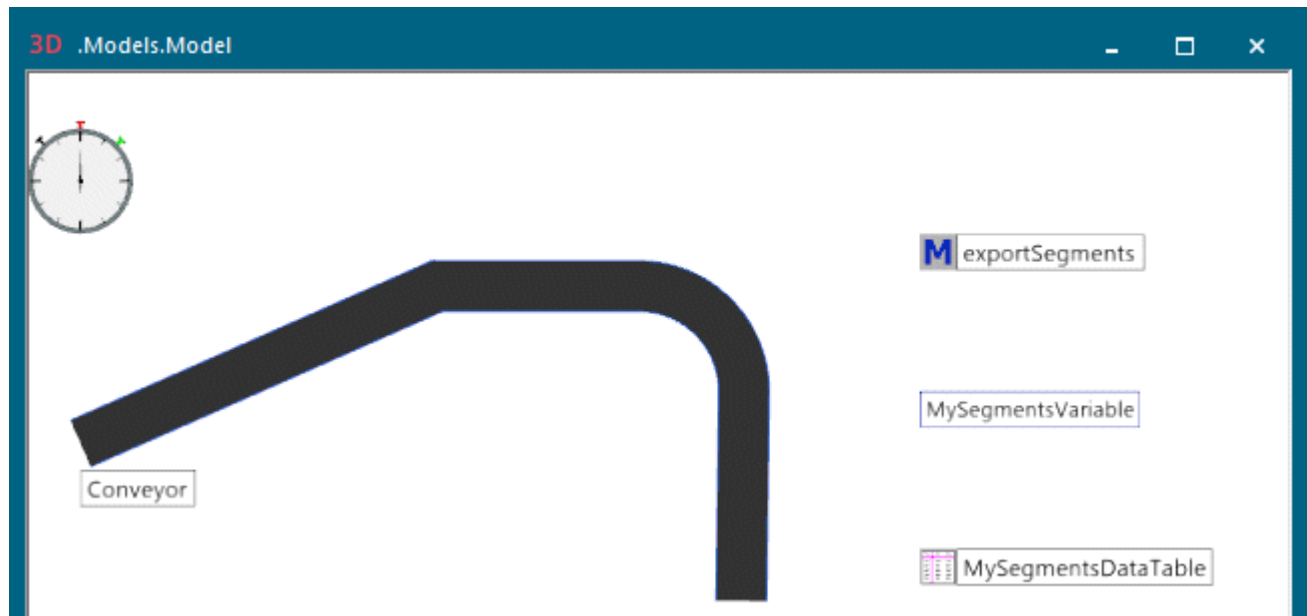
Segmente – .Modelle.Drehtisch.BeschickungTeil2										
I.	Tangentialwinkel [°]	Länge [m]	Kurvenwinkel [°]	Radius [m]	Vertikale Kurve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						-1.11022302462516e-16	-1.668	5.782	1.000	0.000
2	45	2.32625880572562				0	-0.023	4.137	1.000	2.326
3	0		-45	2	☐	0	1.391	3.552	1.000	3.897
4	0	4.60867324735299				0	6.000	3.552	1.000	8.506

We copied the values of the individual cells in the German table and pasted them into the respective cell in the English table. As the English table originally only has three rows, we clicked the right mouse button in the index of row 2 and selected **Append Row**. This appended an empty row below row 2. Then we copied and pasted the remaining values and clicked **OK**.



Segments – .Models.MyTurntable.FeederPartB										
I.	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						-1.11022302462516e-16	-3.750	6.300	1.000	0.000
2	45	2.82842712				0	-1.750	4.300	1.000	2.828
3	0		-45	2	☐	0	-0.336	3.714	1.000	4.399
4	0	4.60867324735299				0	4.273	3.714	1.000	9.008

To transfer the segments table of a curved object from one *Frame* to another *Frame* in the same **simulation model**, we write the segments table into a *Variable* with a *Method*. Then, we copy this *Variable* and paste it into the other *Frame*.

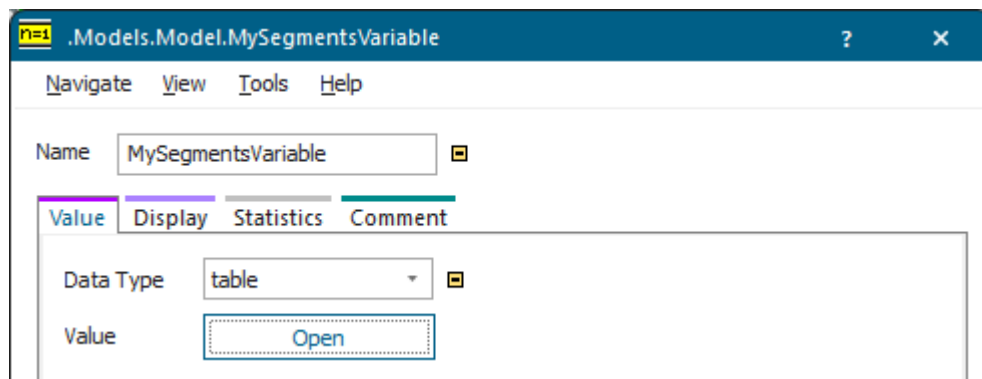


To export the segments table:

- Program a *Method* to export the coordinates of the curved object into a table with the method [getCurveSegments \[SimTalk\]](#). Enter the following source code into the method which we named *exportMySegments*:

```
Conveyor.getCurveSegments(MySegmentsVariable)
```

Executing the method, *exportMySegments* in our example, writes this data into the *Variable* named *MySegmentsVariable*.



Clicking **Open** opens the segments table. Here you can view and check the values of the segments table.

	real 1	length[m] 2	real 3	length[m] 4	boolean 5	length[m] 6	length[m] 7	length[m] 8	length[m] 9	length[m] 10
string	Tangential angle	Length	Curve angle	Radius	Vertical curve	ΔZ	X	Y	Z	ΔL
1						0	1	-6	1	0
2	-23.81	7.4323616027832				0	7.79999993811348	-3.000000027...	1	7.4323616027832
3	23.81	3.85				0	11.6499999381135	-3.000000027...	1	11.2823616027832
4	0.00		90.00	2	false	0	13.6499999381135	-5.000000027...	1	14.423954256373
5	0.72	4.00031242370606				0	13.5999999389145	-8.999999963...	1	18.4242666800791

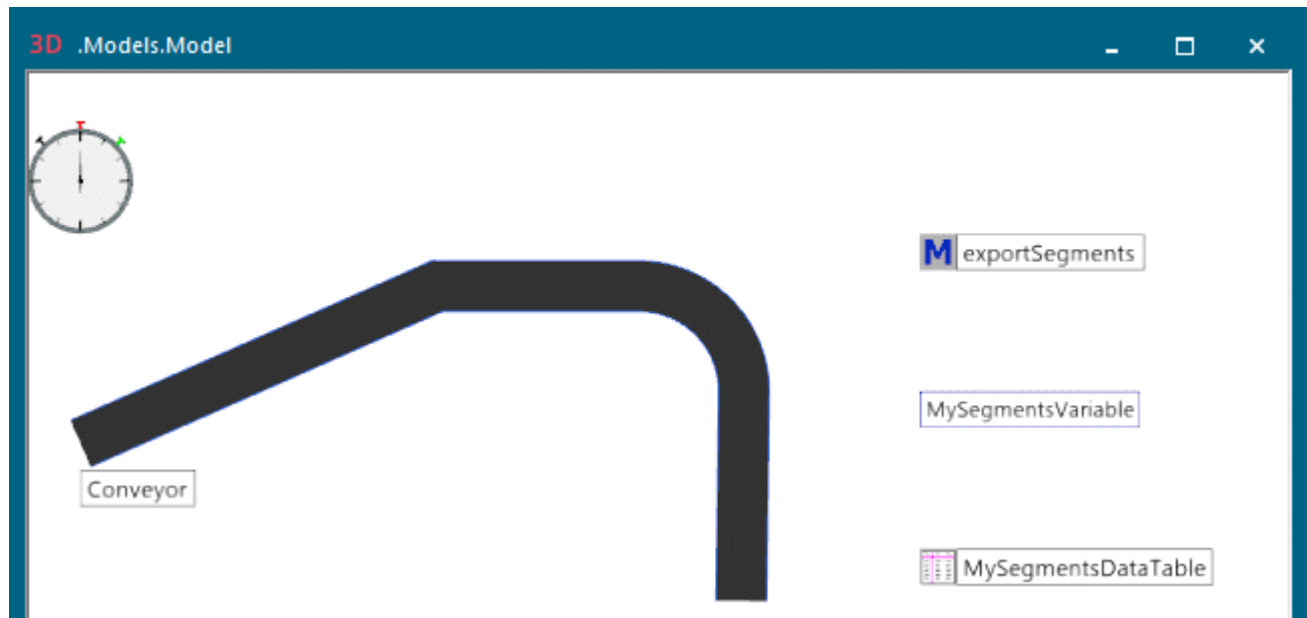
To import the segments table into another *Frame* in the same simulation model:

- Insert a *Conveyor* of any shape into the other *Frame*.
- Copy the *Variable*, *MySegmentsVariable* in our example, and paste it into that *Frame*, *Model1* in our example. Then, insert a *Method* into *Model1*. We named our method *importSegments*.
- Program the *Method* with the method [setCurveSegments \[SimTalk\]](#) to overwrite the settings of the *Conveyor*, which we inserted, with the exported settings. Run the *Method*.

```
Conveyor.setCurveSegments(MySegmentsVariable)
```



If you want to use these settings in **another simulation model**, insert a *DataTable* into that model. **Save** the segments table, *MySegmentsVariable* in our example, **as an object file** (.psobj). You can then import the exported segments into a *DataTable* in the other simulation model and reuse it there.



Back to [Work with Length-oriented Objects](#)

Keyboard Shortcuts for Inserting Straight and Curved Segments of an Object

These keyboard shortcuts apply when you hold down the **Ctrl** key to activate **Curve** mode, drag the mouse and click the mouse button to insert a curved segment.

To do this	Press/click
Set the starting point of a straight segment.	left mouse button
Set the starting point of a curved segment.	Ctrl +left mouse button
Deactivate the Snap To Grid feature.	Alt +left mouse button
Deactivate the Snap To Grid feature for the last point.	Alt +right mouse button
Draw a straight segment horizontally or vertically and to deactivate Snap To Grid .	Shift +left mouse button
Terminate Insert mode.	right mouse button
Insert a new curved object of the same class without having to select it in the <i>Toolbox</i> again.	Ctrl +right mouse button, then left mouse button
Draw the last straight segment horizontally or vertically and to deactivate Snap To Grid .	Shift +right mouse button
The right mouse button just refers to the last anchor point.	

To do this	Press/click
Deactivate default settings	Shift+Alt
Insert the icon of the length-oriented object or of the length-oriented object as a curved object against the default values.	Shift +first click of the left mouse button
Open the dialog Sensor .	Alt +double-click the red Sensor line on <i>Conveyor</i> , <i>Track</i> , <i>TwoLaneTrack</i>
3D : Insert ascending curve	Press the Up key, then the left mouse button
3D : Insert descending curve	Press the Down key, then the left mouse button

Compare Edit Length-oriented Objects with Keyboard Keys

Edit Length-oriented Objects with the Mouse

You can edit the shape and the dimensions of the length-oriented objects and of the *Store* with the mouse.

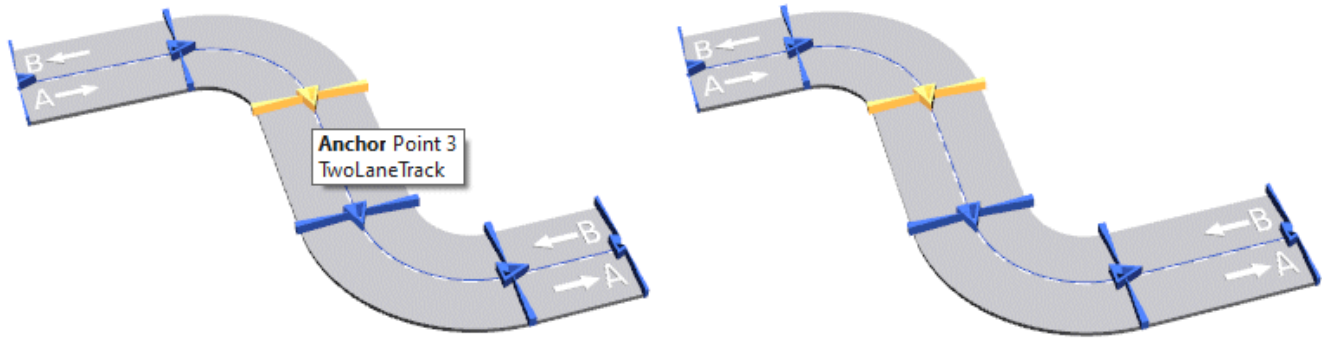
To do so, click **Show Manipulators**  on the **Edit** ribbon tab. This shows all manipulators of the objects. Then, click one of the manipulators with the left mouse button and drag the mouse to the new position and release the mouse button.

The appearance of the manipulator differs from object to object. For the length-oriented objects the manipulator at the start and at the end of the object is cut off. When you attach an object of the same type to the other object, the two combined halves result in a complete manipulator again. When you move the mouse over a manipulator, *Plant Simulation* shows a *tooltip*.

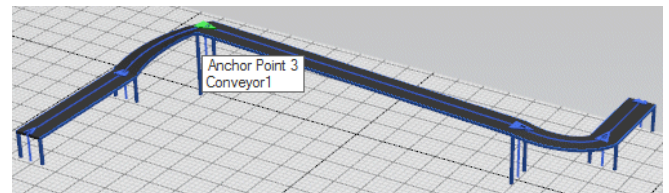
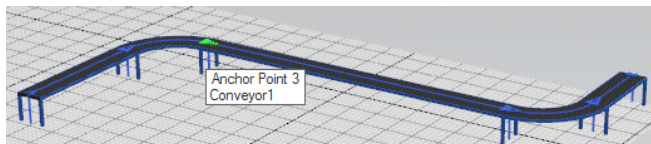


These examples show what you can accomplish with the manipulators:

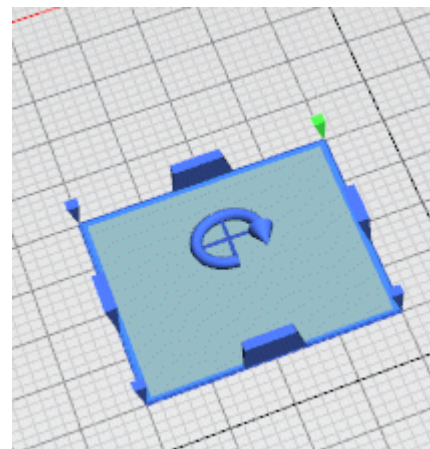
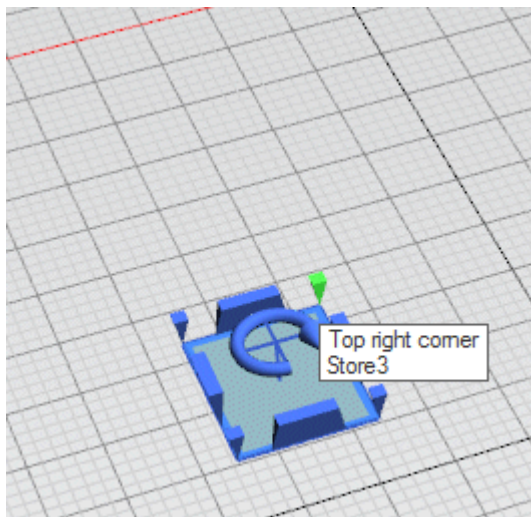
- Here we selected **Anchor Point 3** of the *TwoLaneTrack* and dragged the mouse to the left to shorten the first segment.



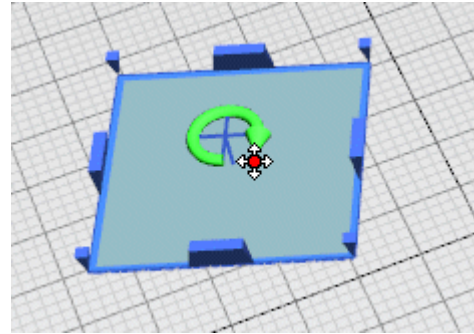
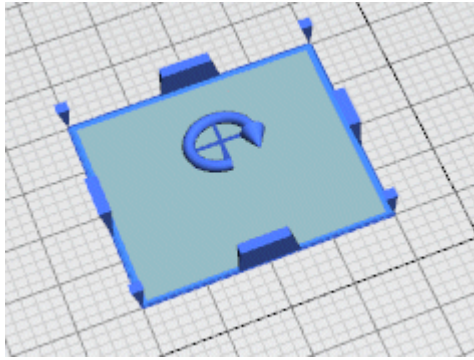
- Here we selected **Anchor Point 3** of the *Conveyor*, we held down **Ctrl** and pressed the **Up** arrow. This moves the anchor point and thus this part of the *Conveyor* up. This way you can raise individual segments, for example to allow *Workers* to walk underneath them or to allow forklifts to pass underneath. Hold down **Ctrl** and the **Up** arrow repeatedly to move the anchor point step-by-step.



- Here we selected the **Top right corner** of the *Store* and dragged the mouse to the top right to enlarge the graphic of the *Store*. The changed size is reflected in the **width** and the **depth** of the *Store* and in the **length** and the **width** of the **animation area**.



To rotate the *Store* around its center, click the rotation manipulator and drag the mouse to the left or to the right.



Back to [Work with Length-oriented Objects](#)

Video on YouTube

<https://youtu.be/Kznlnw5BJGc?si=jK3z2M42kfyRVhUp&t=165>

Editing Length-oriented Objects with Keyboard Keys

Edit Length-oriented Objects with Keyboard Keys

You can edit length-oriented objects with keyboard keys.

You can:

- [Edit Manipulators with Keyboard Keys](#)
- [Move Sensors with Keyboard Keys](#)

Back to [Work with Length-oriented Objects](#)

Compare [Keyboard Shortcuts for Inserting Straight and Curved Segments of an Object](#)

Compare [Move an Object With Keyboard Keys](#)

Edit Manipulators with Keyboard Keys

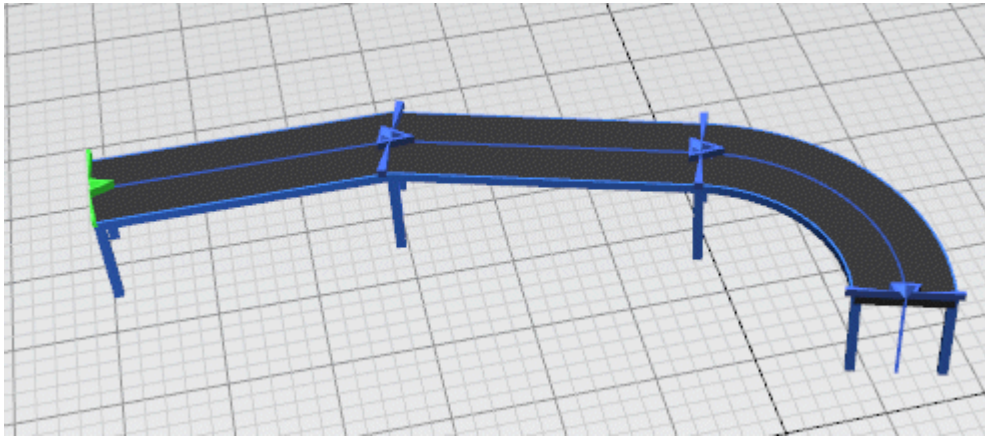
You can use **Page Up** to lengthen the selected manipulator and to shorten it with the **Page Down**.

- Select the **first manipulator of a length-oriented object** and press **Page Up** to lengthen the neighboring segment toward this manipulator.

Page Up lengthens a linear segment in a linear way.

It increases the curve angle for a horizontal curved segment.

It has no effect on a vertical curved segment.

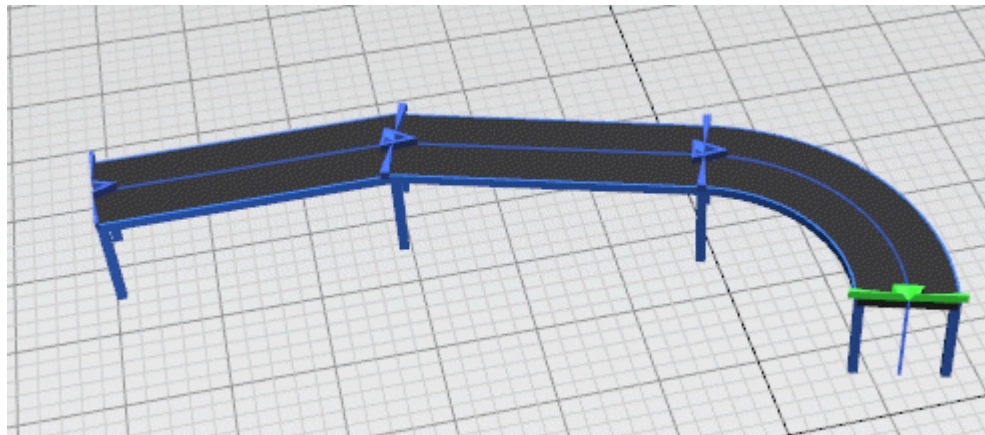


Select the **last manipulator of a length-oriented object** and press **Page Up** to shorten the neighboring segment toward this manipulator.

Page Up shortens a linear segment in a linear way.

It decreases the curve angle for a horizontal curved segment.

It has no effect on a horizontal curved segment.



- Select the **first manipulator of a length-oriented object** and press **Page Down** to shorten the neighboring segment toward this manipulator.

Page Down shortens a linear segment in a linear way.

It decreases the curve angle for a horizontal curved segment. It has no effect on a vertical curved segment.

Select the **last manipulator of a length-oriented object** and press **Page Down** and to lengthen the neighboring segment toward this manipulator.

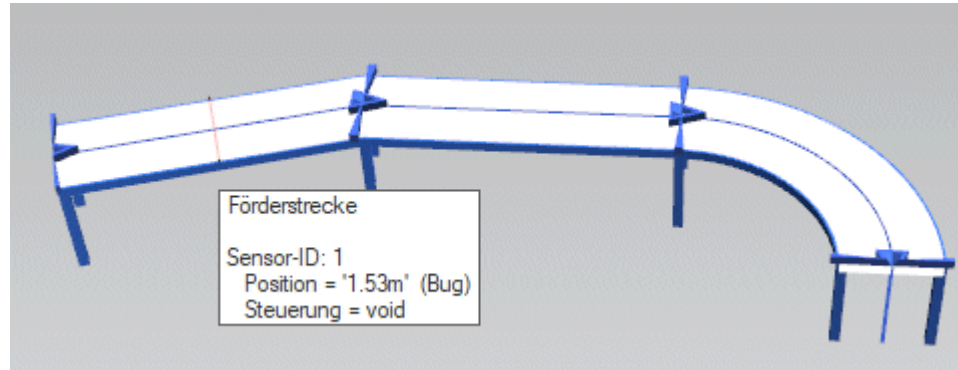
Page Down lengthens a linear segment. It increases the curve angle for a horizontal curved segment. It has no effect on a vertical curved segment.

Back to [Edit Length-oriented Objects with Keyboard Keys](#)

Move Sensors with Keyboard Keys

You can use **Page Up** and **Page Down** to move sensors.

- Select a **sensor** and press **Page Up** to move the sensor to the left. The **left arrow** does the same.
- Select a **sensor** and press the **Page Down** to move the sensor to the right. The **right arrow** does the same.



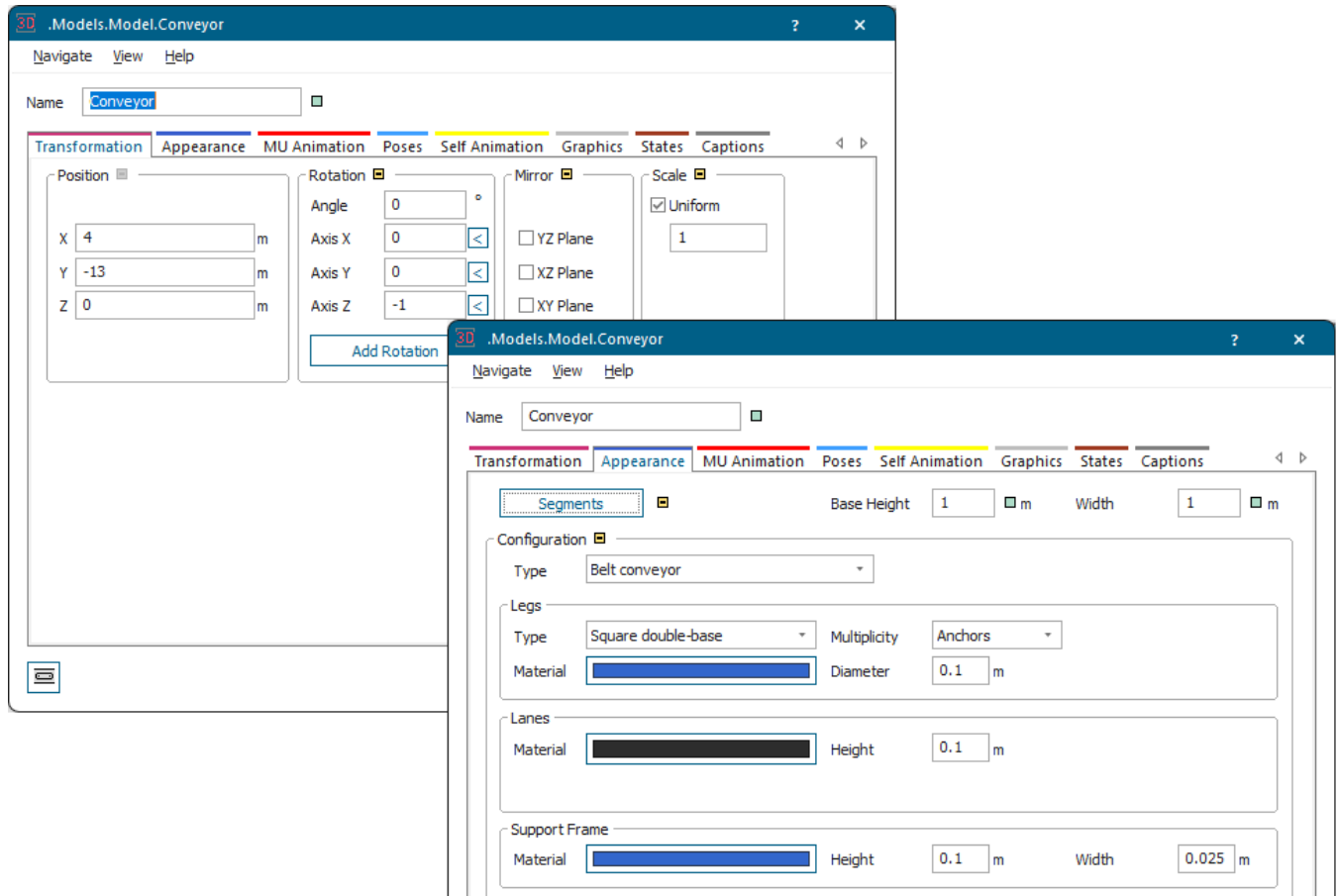
Back to [Edit Length-oriented Objects with Keyboard Keys](#)

Edit Length-oriented Objects Precisely

You can change the length-oriented objects in your simulation model precisely by entering exact values into the text boxes into a dialog.

For some purposes this is not accurate enough though. If this is the case, you can edit the **position**, the **rotation**, and the **scaling** of the object in the dialog **Edit 3D Properties** on the [Tab Transformation](#) and on the tab [Appearance of the Length-oriented Objects](#), by entering exact values into the text boxes.

- Select the object that you want to edit.
- Press the spacebar or click  on the **Home** ribbon tab and click the tab **Transformation**.
- Edit the values of the **position**, the **rotation**, and the **scaling**.
- Click **Apply** to apply your changes once you are satisfied with the results.
- If necessary, you can edit additional settings on the tab **Appearance**.



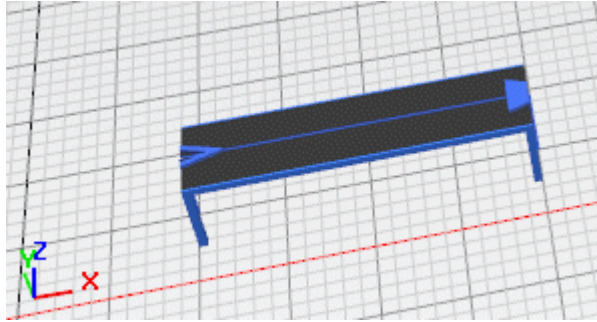
Back to [Work with Length-oriented Objects](#)

Model Differences in Height Between Conveying Elements

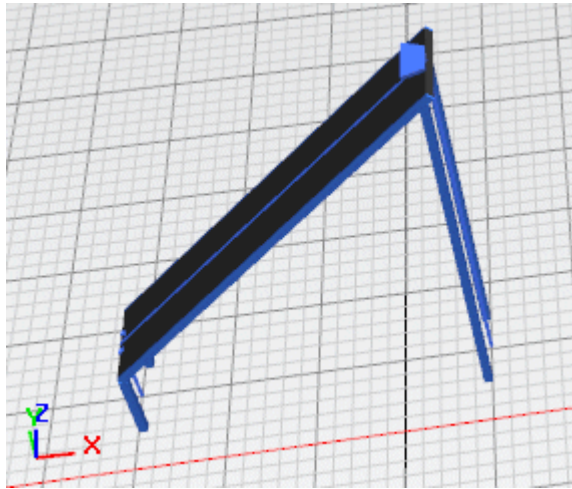
With the length-oriented objects you can create straight and curved ascending and descending slopes in your simulation model to model height differences between them.

Proceed as follows to model **straight ascending and descending slopes**:

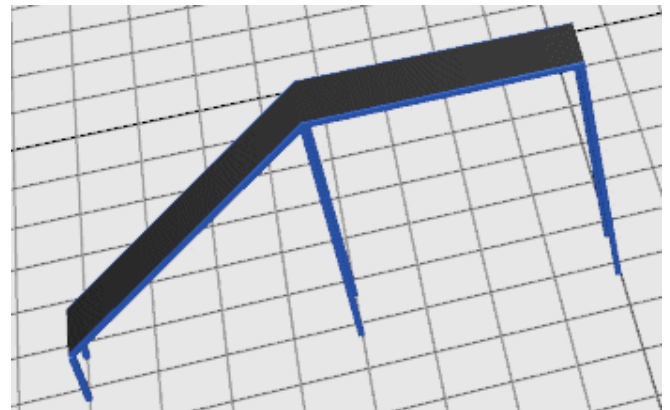
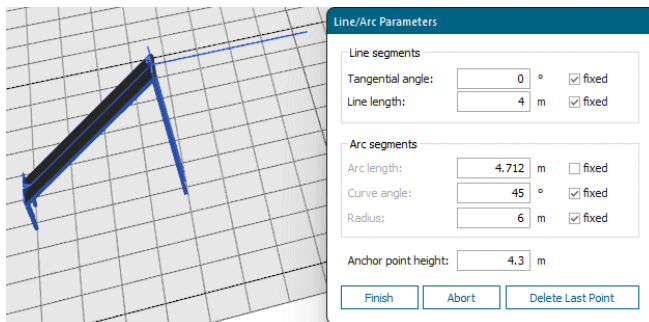
- Select the length-oriented object that you want to insert into the *Frame*. In our example we insert a *Conveyor* that is 4 meters long. Then, click **Show Manipulators** on the **3D** ribbon tab to show the manipulators.



- Click the right manipulator with the right mouse button, hold down **Ctrl** and press the **Up** arrow until the right side of the *Conveyor* has the required height.



- To append a segment to the ascending slope, click the *Conveyor* with the right mouse button and select **Segments > Append**.
- Type in the radius and the angle of the segment into the dialog **Line/Arc Parameters**. Click in the scene to append the segment. Click **Finish** to close the dialog.



Proceed as follows to model **curved ascending and descending slopes**:

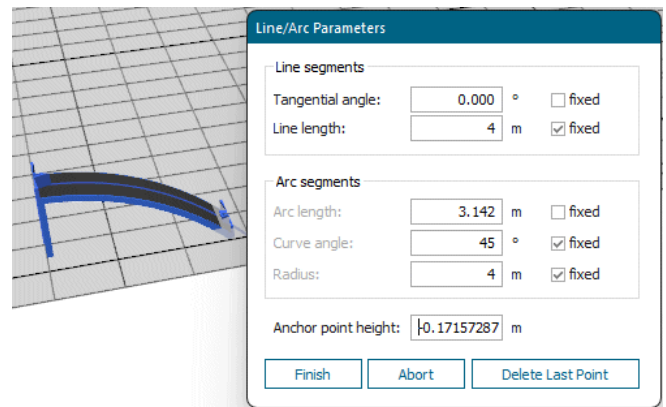
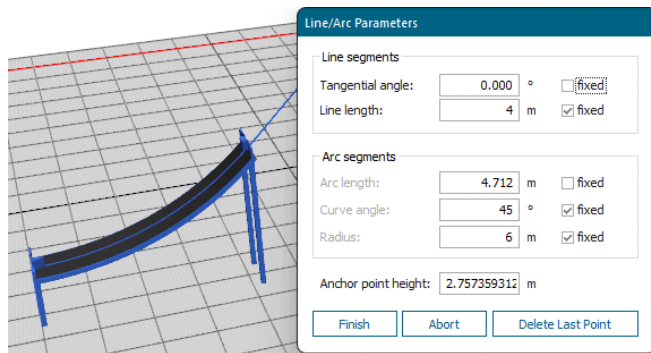
- Select the length-oriented object that you want to insert into the *Frame*. In our example we insert a *Conveyor*.
- Press the **Up arrow** on the keyboard to insert an ascending slope, a curve pointing upward in the **z-dimension**.

Press the **Down arrow** on the keyboard to insert a descending slope, a curve pointing downward in the **z-dimension**.

- Type in the **Radius** and the **Angle** of the segment into the dialog **Line/Arc Parameters**.

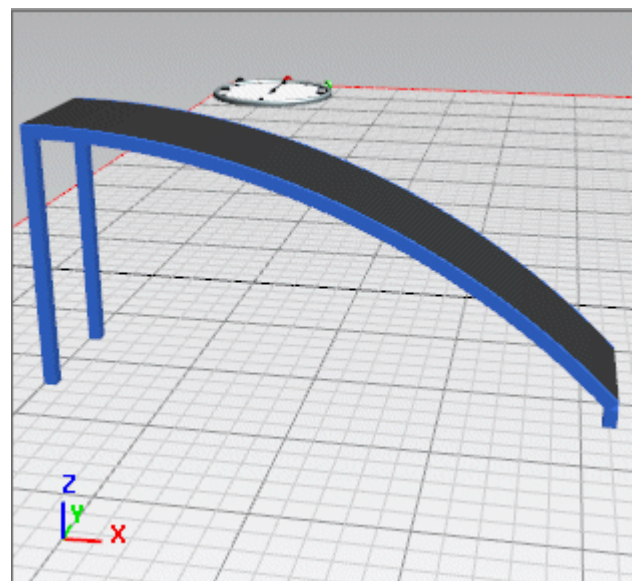
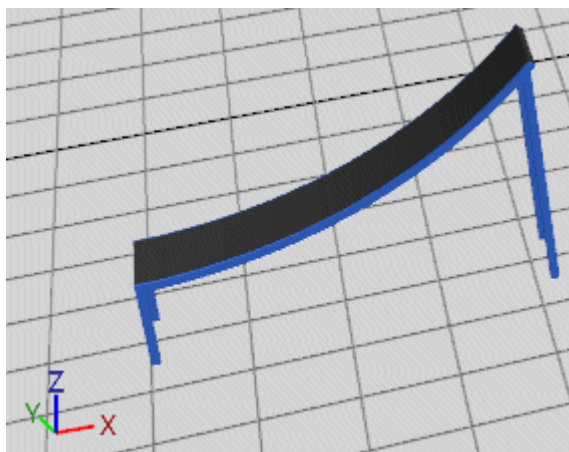
Note

The greatest allowed **curve angle** is 90°.

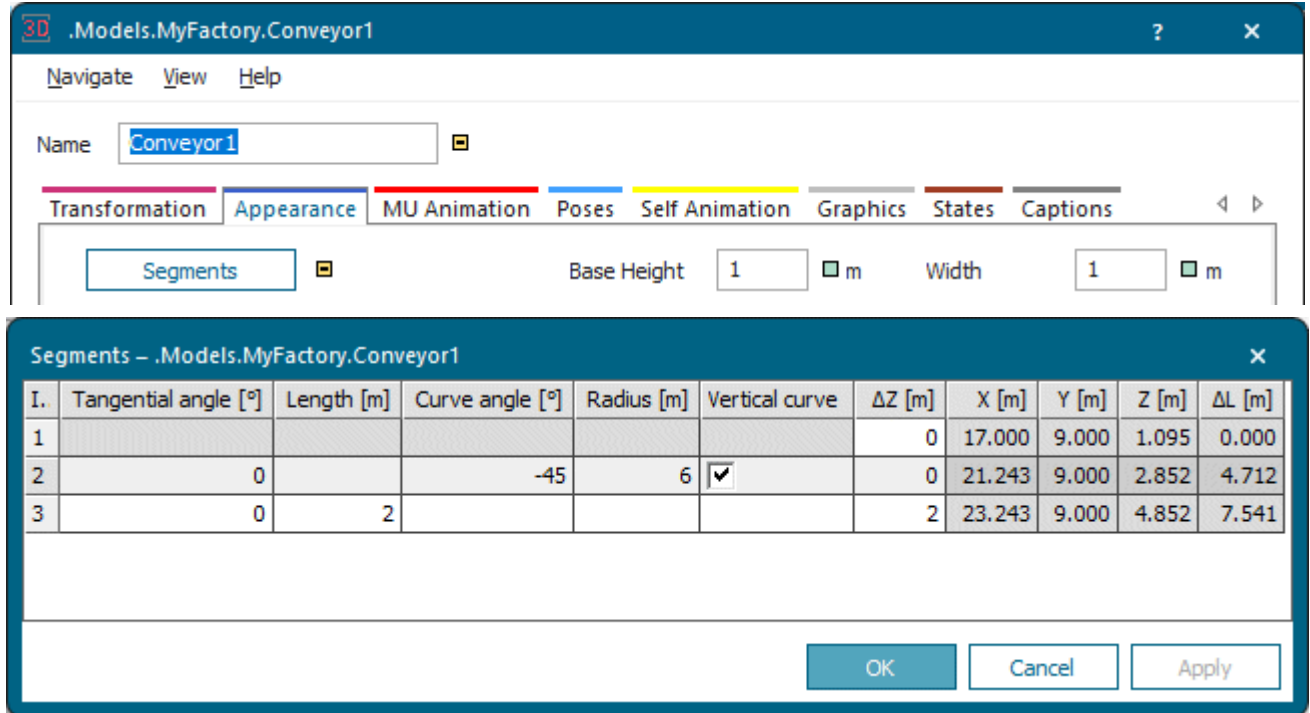


- Click with the mouse into the scene to insert the segment. Click **Finish** to close the dialog.

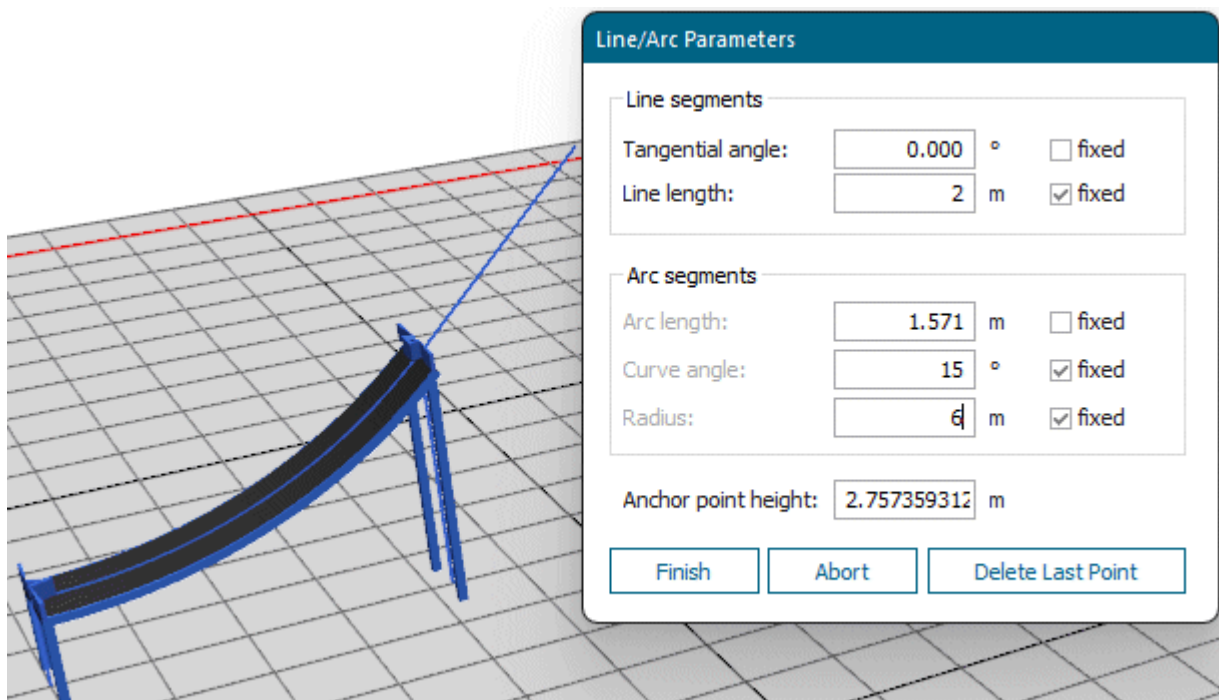
The result looks like this:



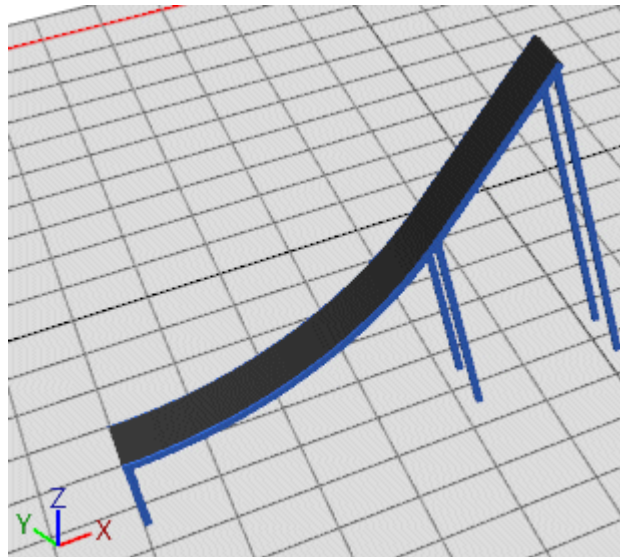
When you click the **Segments** button, the selected check box in the dialog shows, that our *Conveyor* contains a **vertical curve**.



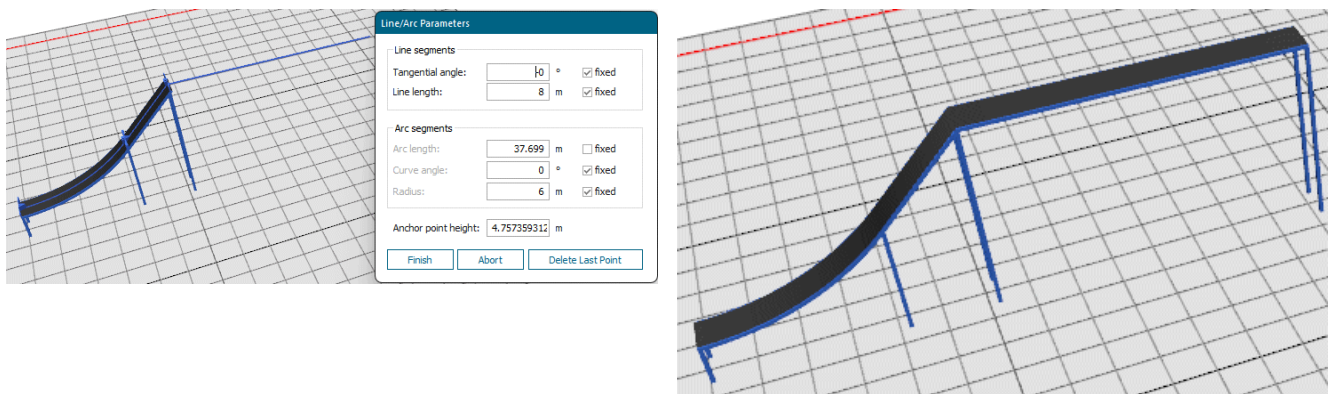
- To append segments to the ascending slope, click the *Conveyor* with the right mouse button and select **Segments > Append**.
- Type in the **Radius** and the **Angle** of the segment into the dialog **Line/Arc Parameters**. Click the left mouse button to insert the segment you just defined and click **Finish** in the dialog.



- The result looks like this.



- Repeat this until the Conveyor has the shape that you need.



Compare [Keyboard Shortcuts for Inserting Straight and Curved Segments of an Object](#)

Back to [Work with Length-oriented Objects](#)

Model a Spiral Conveyor

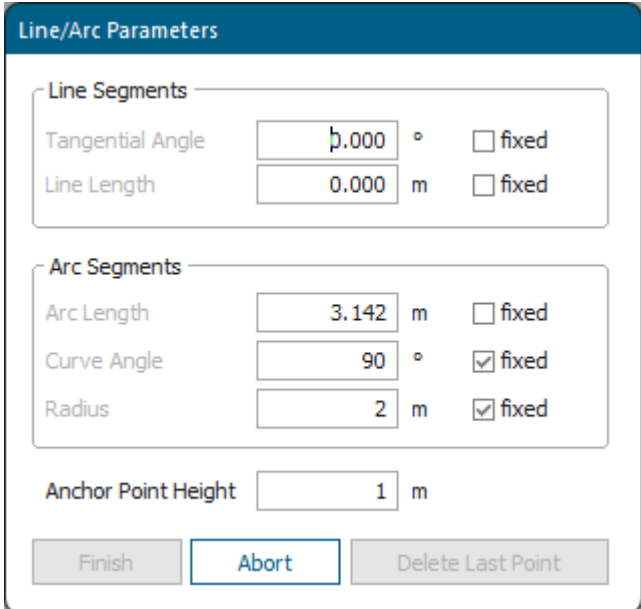
You can model a spiral conveyor in your simulation with the object *Conveyor*.

A spiral conveyor transports products upward or downward thus leveling differences in height within your production facility. Depending on the height difference you can also use the spiral conveyor as a buffer.

Proceed as follows to model the spiral conveyor:

- Click the *Conveyor* on the tab **MaterialFlow** of the **Toolbox**.
- Drag the mouse in the window to the position at which you would like to insert the conveyor.

- We used the standard settings in the dialog **Line/Arc Parameters**. If you need to use a wider arc, enter a greater **radius**.



The dialog box titled "Line/Arc Parameters" contains two main sections: "Line Segments" and "Arc Segments".

Line Segments:

- Tangential Angle: 0.000 ° ☐ fixed
- Line Length: 0.000 m ☐ fixed

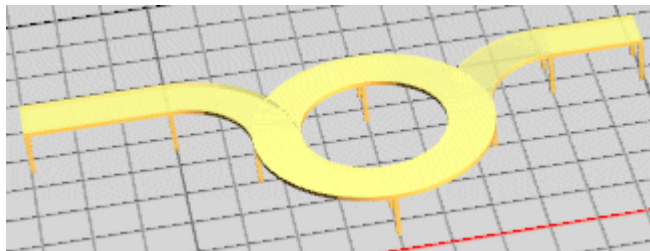
Arc Segments:

- Arc Length: 3.142 m ☐ fixed
- Curve Angle: 90 ° ☒ fixed
- Radius: 2 m ☒ fixed

Anchor Point Height: 1 m

Buttons at the bottom: Finish, Abort, Delete Last Point.

- Hold down **Ctrl** to insert the arc segments of the spiral conveyors. Click several times one after the other to insert the arc segments. The segments are initially placed on a single layer directly above each other. We first inserted a straight segment, then added 13 arc segments, and finished with a straight segment.



- Press **F7** to open the **Segments** table.

Segments – .Models.MyFactory.Conveyor3

I...	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	17.000	5.000	1.000	0.000
2	0	3				0	20.000	5.000	1.000	3.000
3	0		90	2	☐	0	22.000	3.000	1.000	6.142
4	0		-90	2	☐	0	24.000	1.000	1.000	9.283
5	0		-90	2	☐	0	26.000	3.000	1.000	12.425
6	0		-90	2	☐	0	24.000	5.000	1.000	15.566
7	0		-90	2	☐	0	22.000	3.000	1.000	18.708
8	0		-90	2	☐	0	24.000	1.000	1.000	21.850
9	0		-90	2	☐	0	26.000	3.000	1.000	24.991
10	0		-90	2	☐	0	24.000	5.000	1.000	28.133
11	0		-90	2	☐	0	22.000	3.000	1.000	31.274
12	0		-90	2	☐	0	24.000	1.000	1.000	34.416
13	0		-90	2	☐	0	26.000	3.000	1.000	37.558
14	0		90	2	☐	0	28.000	5.000	1.000	40.699
15	0	2				0	30.000	5.000	1.000	42.699

OK Cancel Apply

- Enter the additional offset to the base height of the preceding segment into the cell ΔZ . We typed in 0.5 meters.

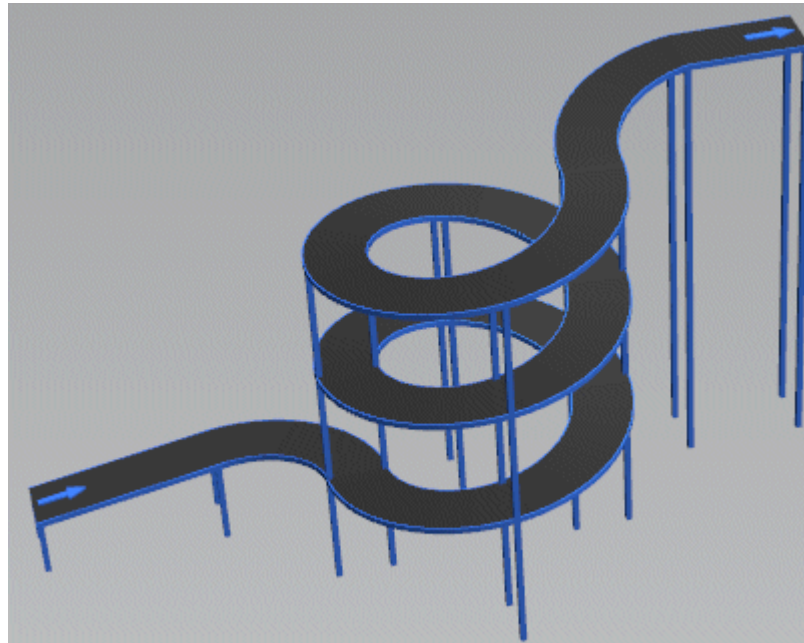
Segments – .Models.MyFactory.Conveyor3

I...	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	17.000	5.000	1.000	0.000
2	0	3				0.5	20.000	5.000	1.000	3.000
3	0		90	2	☐	0.5	22.000	3.000	1.000	6.142
4	0		-90	2	☐	0.5	24.000	1.000	1.000	9.283
5	0		-90	2	☐	0.5	26.000	3.000	1.000	12.425
6	0		-90	2	☐	0.5	24.000	5.000	1.000	15.566
7	0		-90	2	☐	0.5	22.000	3.000	1.000	18.708
8	0		-90	2	☐	0.5	24.000	1.000	1.000	21.850
9	0		-90	2	☐	0.5	26.000	3.000	1.000	24.991
10	0		-90	2	☐	0.5	24.000	5.000	1.000	28.133
11	0		-90	2	☐	0.5	22.000	3.000	1.000	31.274
12	0		-90	2	☐	0.5	24.000	1.000	1.000	34.416
13	0		-90	2	☐	0.5	26.000	3.000	1.000	37.558
14	0		90	2	☐	0.5	28.000	5.000	1.000	40.699
15	0	2				0	30.000	5.000	1.000	42.699

OK Cancel Apply

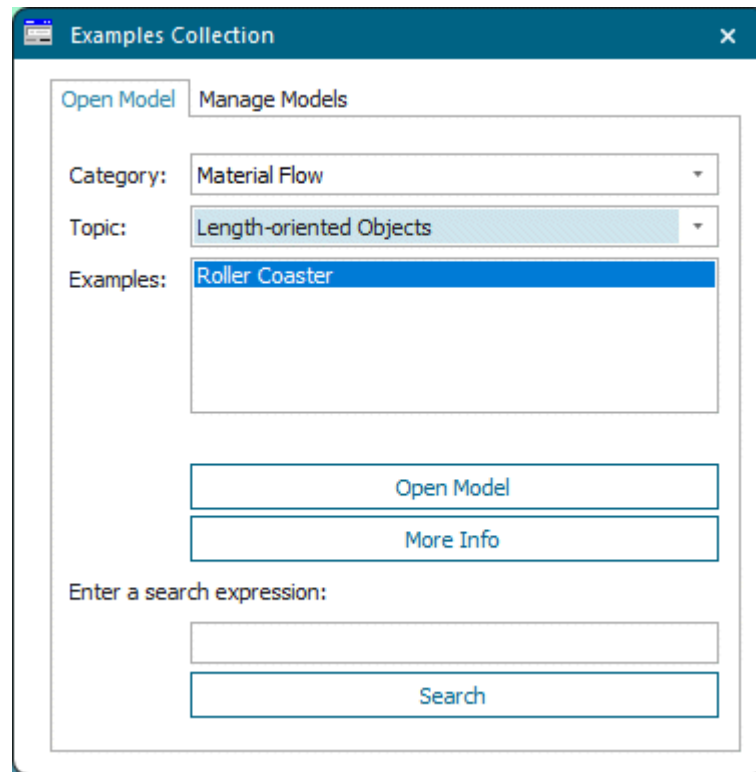
- Click **OK** in the dialog **Segments**.

The result looks like this:



You can change the settings of the spiral conveyor on the tab **Appearance**.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



Back to [Work with Length-oriented Objects](#)

Setting the Capacity of a Material Flow Object

Set the Capacity of a Material Flow Object

While you set the capacity of a material flow object in the simulation properties **in two dimensions**, you set the position of a part on a simulation object in **in three dimensions**.

You can:

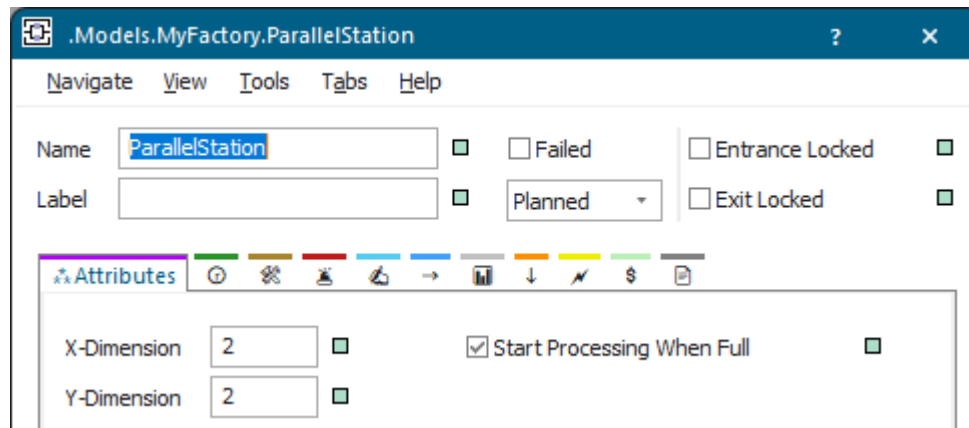
- [Define the Capacity of a Simulation Object](#)
- [Define the Animation Area of a Simulation Object](#)

Back to [Work with Objects](#)

Define the Capacity of a Simulation Object

For the simulation object you define the capacity, i.e., the number of parts the objects *ParallelStation*, *Sorter*, *Store*, *Transporter*, and *Container* can hold, in **two dimensions**.

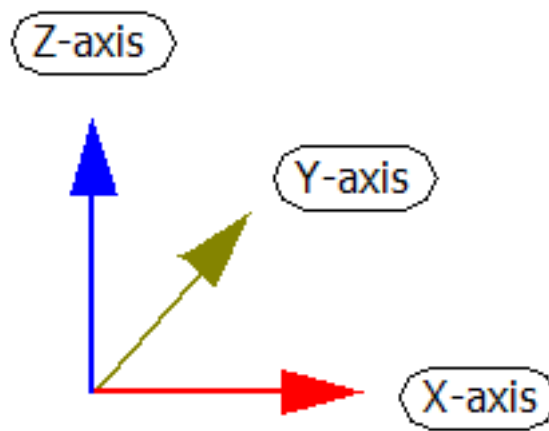
Type the respective values into the text boxes **X-Dimension** and **Y-Dimension** on the tab **Attributes**.



Back to [Set the Capacity of a Material Flow Object](#)

Define the Animation Area of a Simulation Object

You set the animation of the *simulation object* in **three dimensions** on the **animation area**.

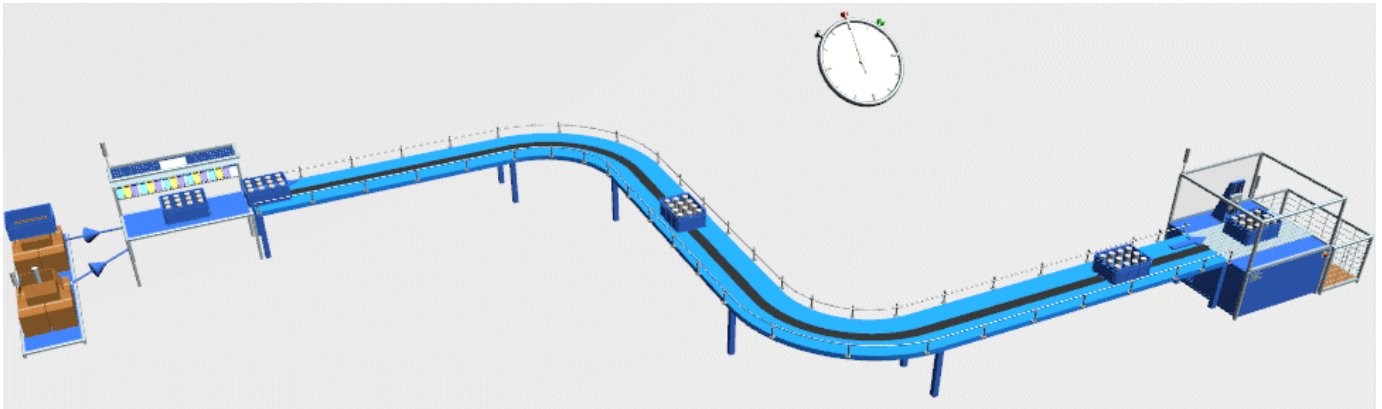


To set the animation area of an object with matrix loading space of type **Store** (*ParallelStation*, *Sorter*, *Store*, *Transporter*, *Container*), of the *PlaceBuffer*, and the *Worker* in 3D, proceed as follows:

- Click the object with the right mouse button to select it.
- Select **Edit 3D Properties > MU Animation > Area > [Animation Area](#)**.

In our example we demonstrate how to place cylinder-shaped *containers* in a *storage box*. This resembles a bottling plant in which the filled bottles are put into cases.

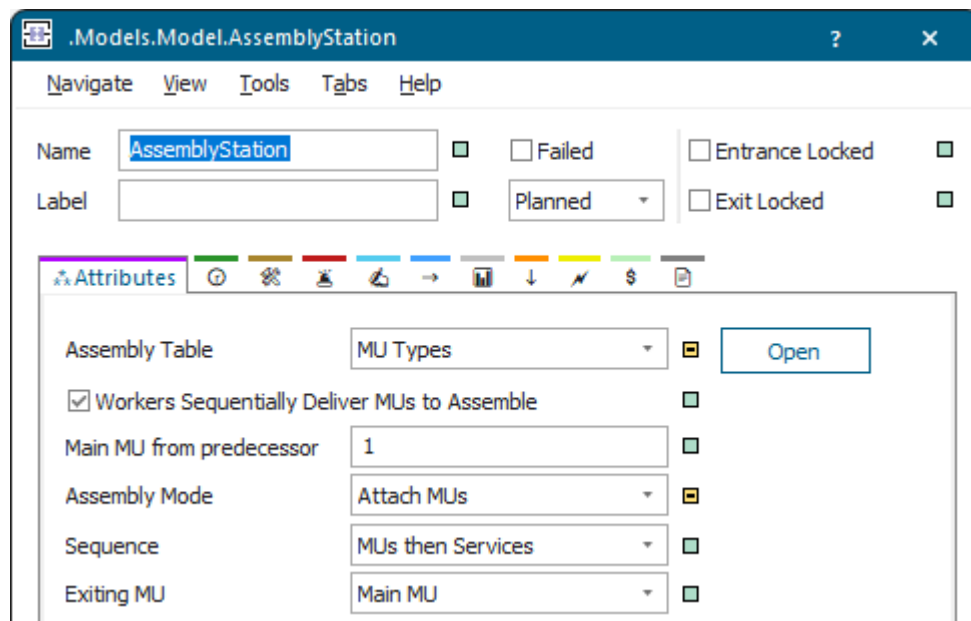
Our finished model looks like this:



A *Source* each produces the *storage boxes* and the cylinder-shaped *containers*. The *AssemblyStation* places the *containers* into the *storage boxes* and then moves them on across a *Conveyor* and a *Station* to the *Drain*.

Proceed as follows to create the simulation model:

- Insert two *Sources*. We named them *SourceStorageBoxes* and *SourceCylinders*.
- Insert an *AssemblyStation*. We selected the settings below:



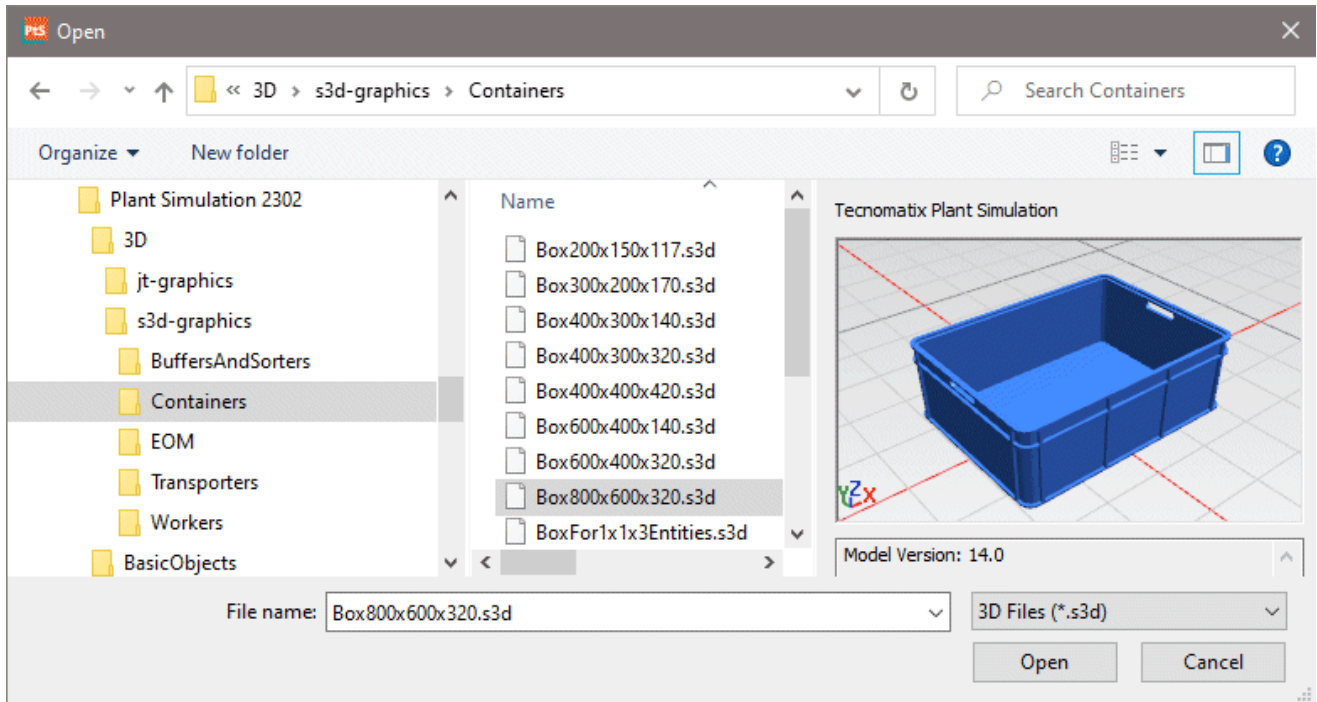
- Insert a *Conveyor*. We used a *Conveyor* of type **Guide-railed belt conveyor**.
- Insert a *Station* and a *Drain*.
- Make sure that all objects are connected with *Connectors*.

As we need customized **MU Types** for the *storage boxes* and the *cylinder-shaped containers*, we duplicate the *Part* and the *Container* in the folder *MUs* in the *Class Library*.

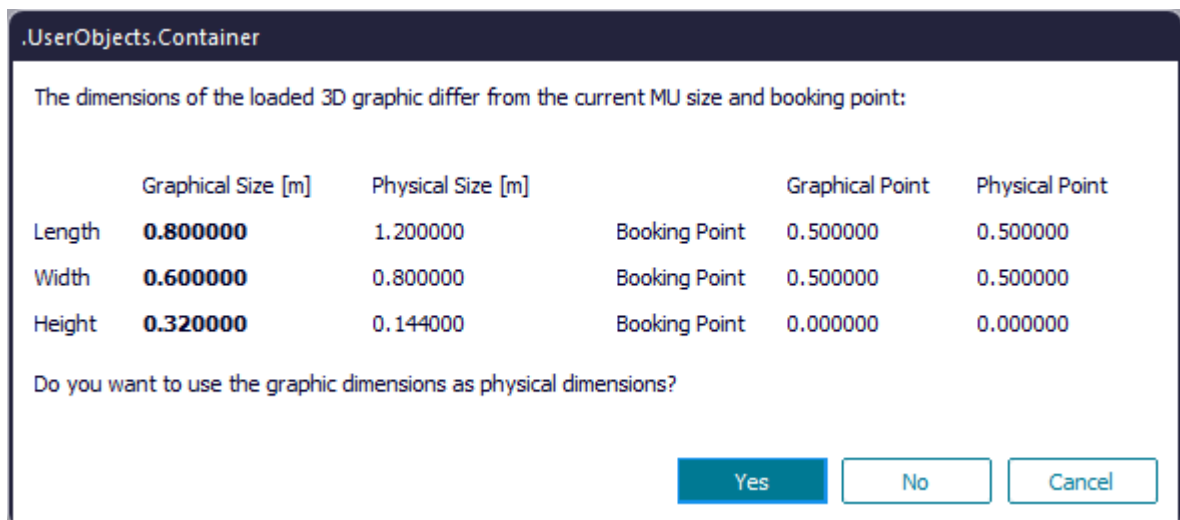
- Press **F2** and rename the duplicated *container* from *Container* to *StorageBox*.

To configure the *storage box*:

- Click the *storage box* with the right mouse button and open it in a window.
- Click with the right mouse button into the background of the window and select **Exchange Graphics**. We selected the *storage box* below.



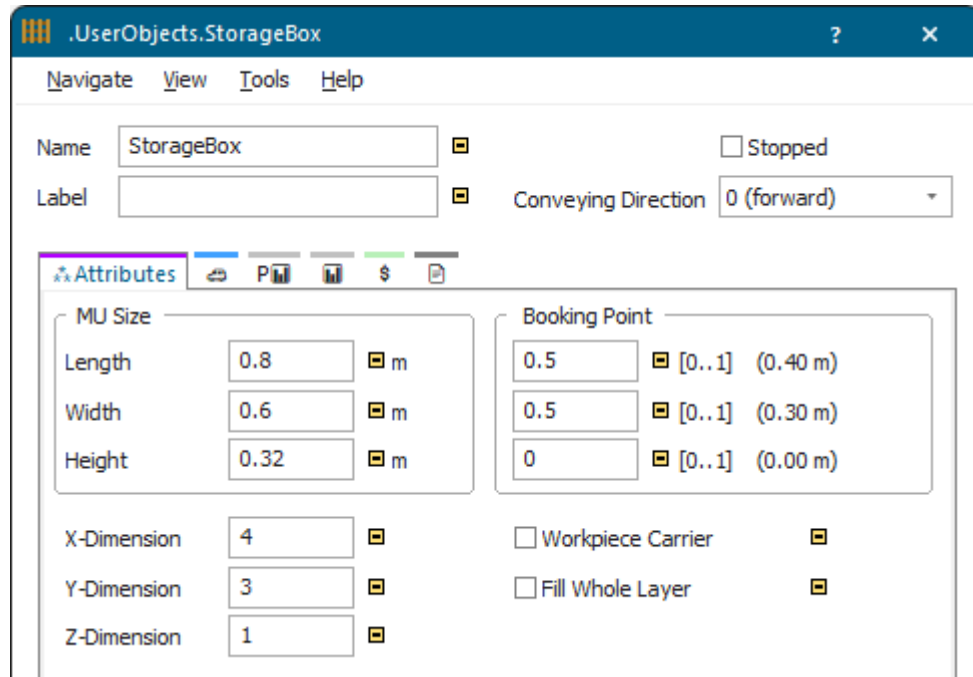
Click **Yes** in the message dialog that *Plant Simulation* shows.



As you remember the *s3D* files do not only contain the graphic of the object, but also information about its dimensions and animation data.

If you click **No**, *Plant Simulation* does not apply the adjustments of the dimensions of the *storage box* and scales it to the current size of the MU named *Container*, as the check box **Scale Automatically** on the tab **Transformation** is selected by default.

- Type in the **dimensions** in the **X-Dimension**, **Y-Dimension**, and **Z-Dimension**, that set its capacity.

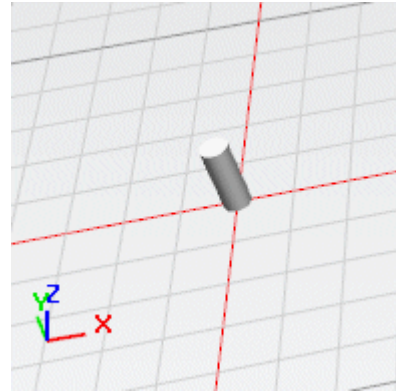
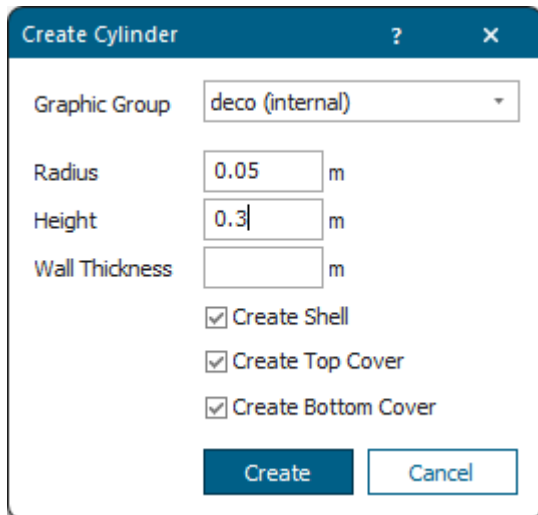


- Press **F2** and rename the duplicated *part* to *Canister*.

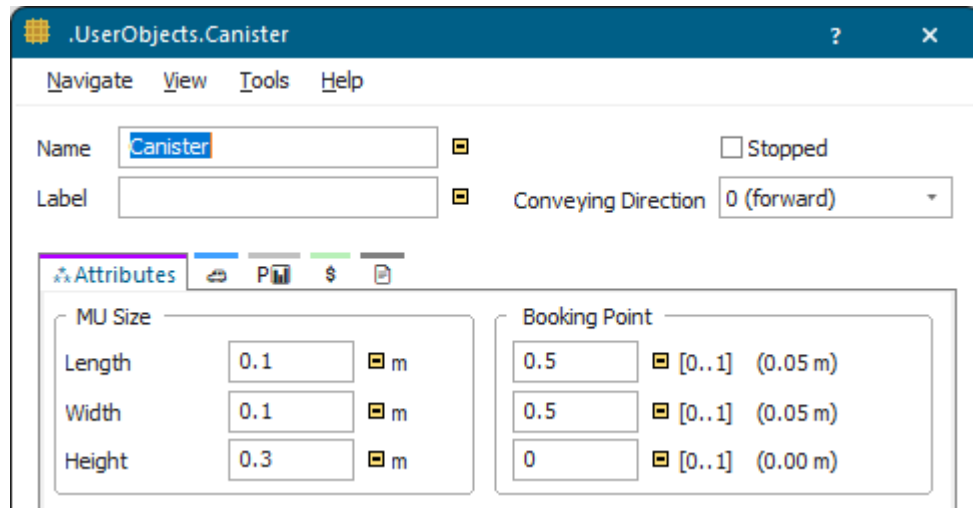
To configure our *Canister*:

- Click the *Canister* with the right mouse button and open it in 3D.
- Click the default graphic of the *Part* and delete it. We are going to create our own graphic for our *Canister*.

Click the **Edit** ribbon tab and click *Cylinder*. Type in the values below. Click **Create**. Make sure that our *Canister* is placed on the origin of the grid.

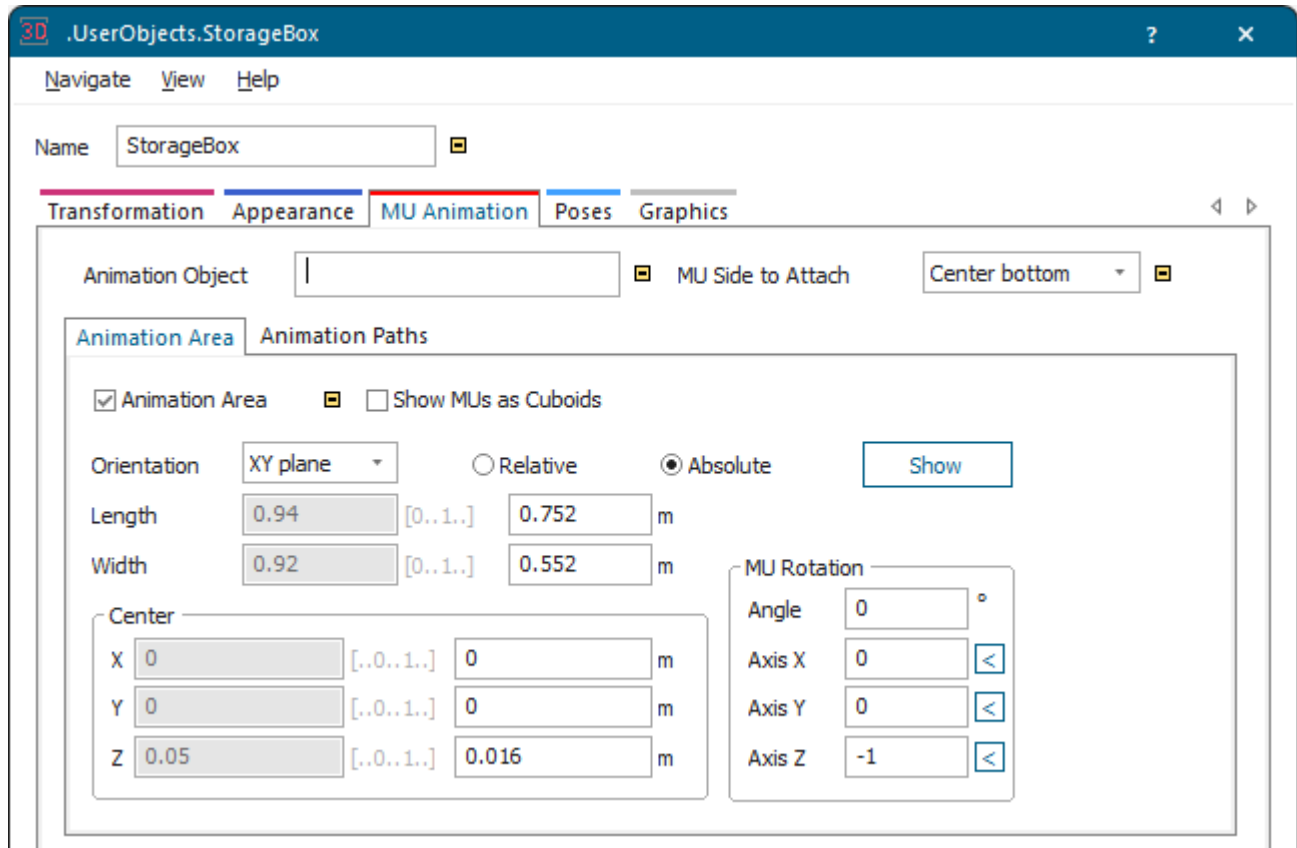


- Now we have to define the **length**, the **width**, and the **height** of our *Canister*. Double-click it in the *Class Library*. Type in these values:

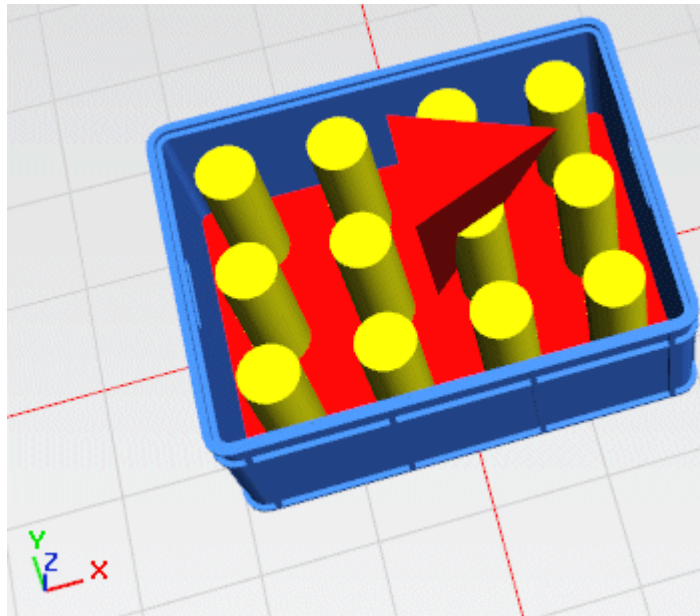


Run the simulation and watch how the *AssemblyStation* places 12 *containers* each in a *storage box* and then moves the full *storage box* on.

The **animation area** of the *storage box*, which is activated by default, makes sure that the *canisters* are distributed evenly across the bottom of the *storage box*. In doing so, *Plant Simulation* automatically uses these settings:



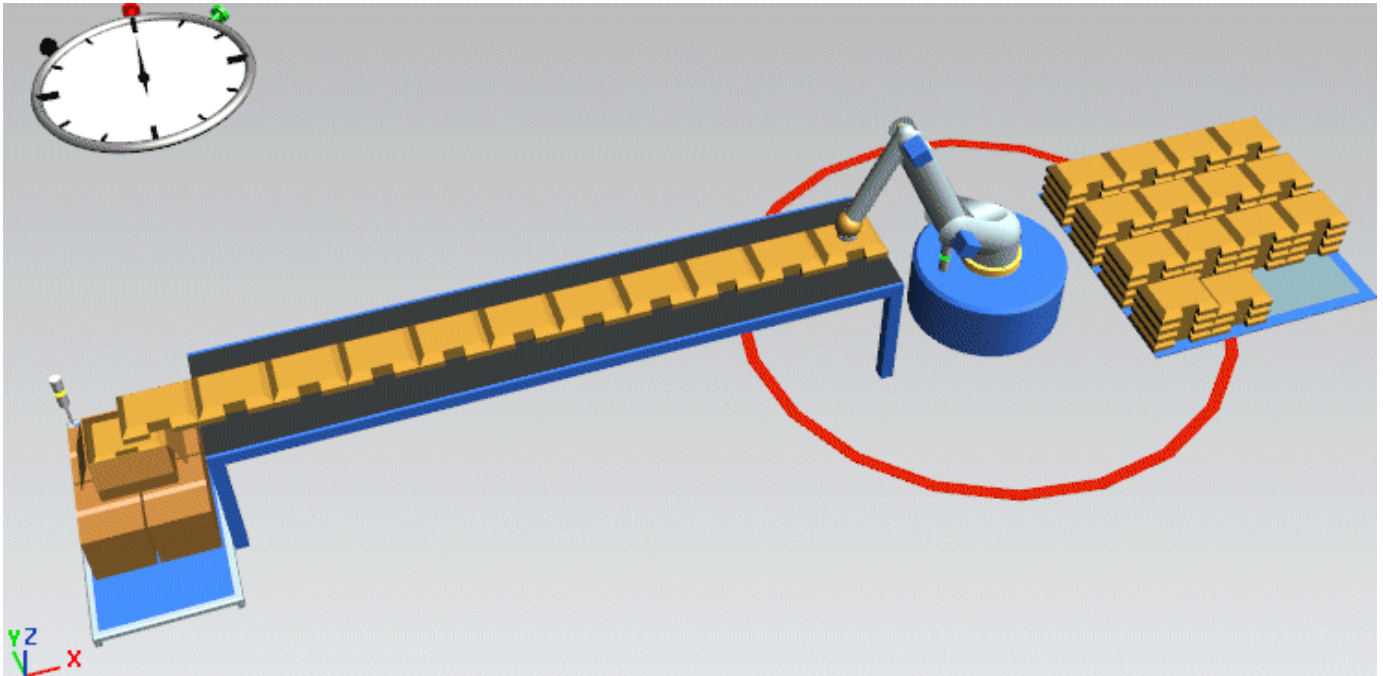
- Click **Show/Hide**, to show the **animation area** in the window to see what it looks like. *Plant Simulation* shows the animation area proper in red with a red orientation indicator in the center.



- The **orientation** > **XY plane** evenly distributes the *containers* in the x-y-direction.

- For the **length** *Plant Simulation* only uses 94 % of the **Length** of the graphic of the *storage box* of 80 cm and for the **width** only 92 % of the **Width** of 60 cm. *Plant Simulation* automatically subtracted the **Wall Thickness** of the *storage box* from the dimensions of the **Animation Area**. *Plant Simulation* uses the area settings that we defined in the s3D graphic.
- For the **Z-Position** of the **center** *Plant Simulation* automatically entered 0.05 m as the bottom of the *storage box* has a height of 5 cm.

Stack Parts in the Store demonstrates how to distribute parts across the storage area and how to stack them.



Compare **Model a Bottle**

Compare **Animate Bricks on the Animation Area of the Brick Carrier**

Compare **Animate Brick Carriers on the Animation Area of the Truck**

Compare **Model a Worker Who Carries Several Parts**

Context Menu of the Background of the Scene

Context Menu of the Background of the Window

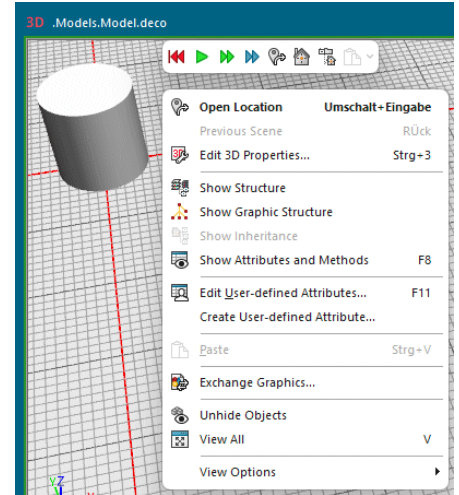
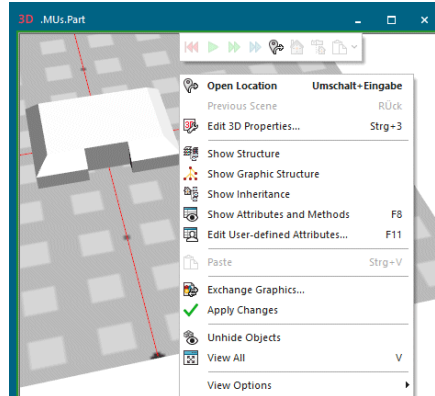
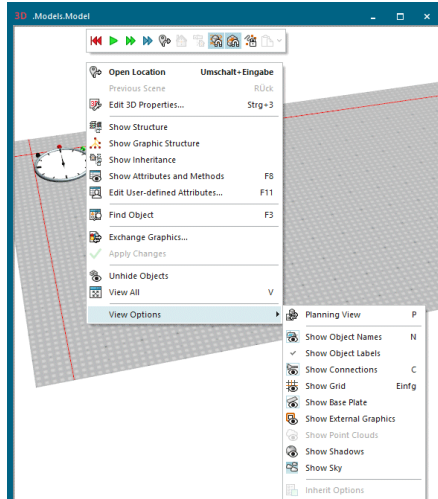
Click in an empty area in the window with the right mouse button to show a context menu with these commands.

You can also close the context menu and leave the *mini toolbar* open



button.

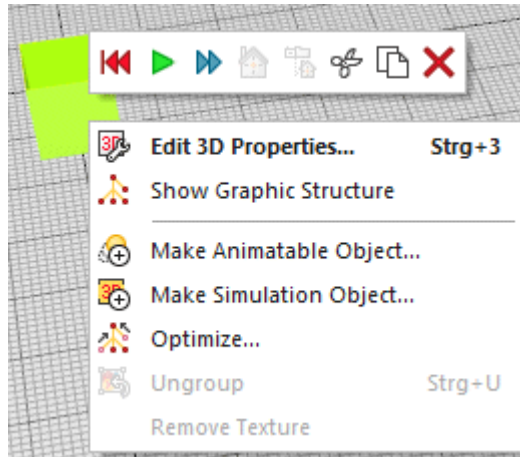
by clicking the border of the *mini toolbar* with the left mouse



 Reset Simulation [Home ribbon]	 Show Structure [Home ribbon]
 Start/Stop Simulation [Home ribbon]	 Show Graphic Structure [context menu]
 Start Fast Forward Simulation [Home ribbon]	 Show Inheritance [Home ribbon]
 Open Location [Home ribbon]	 Show Attributes and Methods [Home ribbon]
 Open Origin [Home ribbon]	 Edit User-defined Attributes [Home ribbon]
 Open Class [Home ribbon]	Create User-defined Attribute
 Snap to Grid	 Find Object [context menu]
 Snap to Objects	 Exchange Graphics
 Lock Structure [Frame ribbon]	 Apply Changes [MUs]
 Paste Contents of the Clipboard [Home ribbon], Multiple Paste	 Unhide Objects

Previous Scene	 View All
 Edit 3D Properties [dialog]	View Options , compare the View Ribbon Tab

Do not confuse the selected object with the selected graphic of the object!



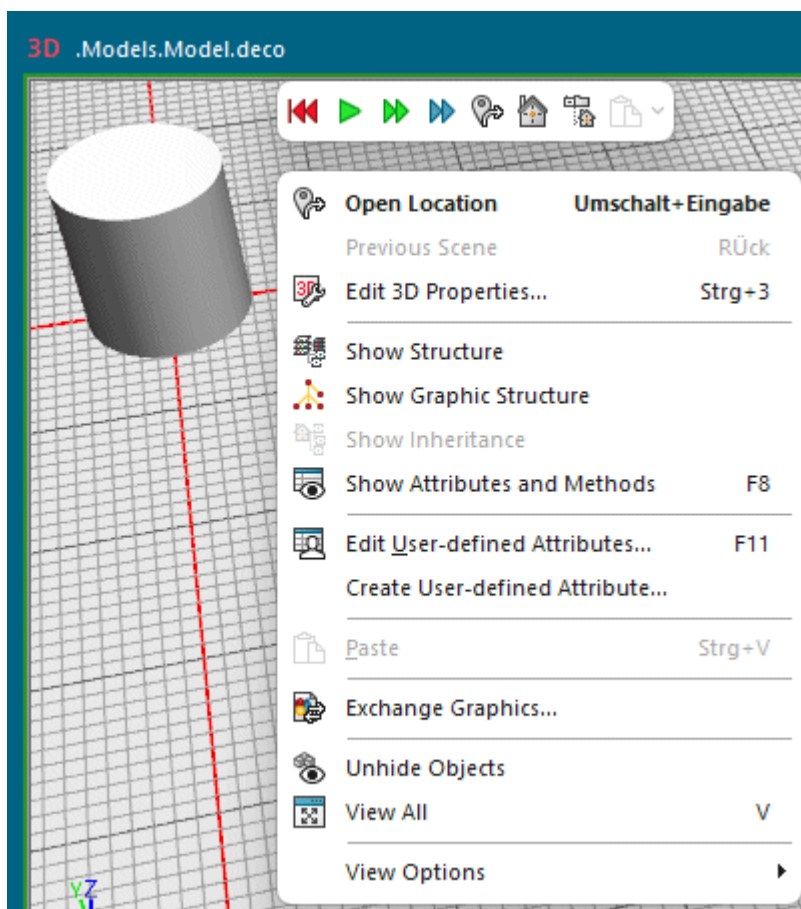
Create User-defined Attribute

To create a *user-defined attribute* for a simulation object, which you opened in a window of its own, select this command.

Plant Simulation creates the *user-defined attribute* at the position, where you click the mouse button.

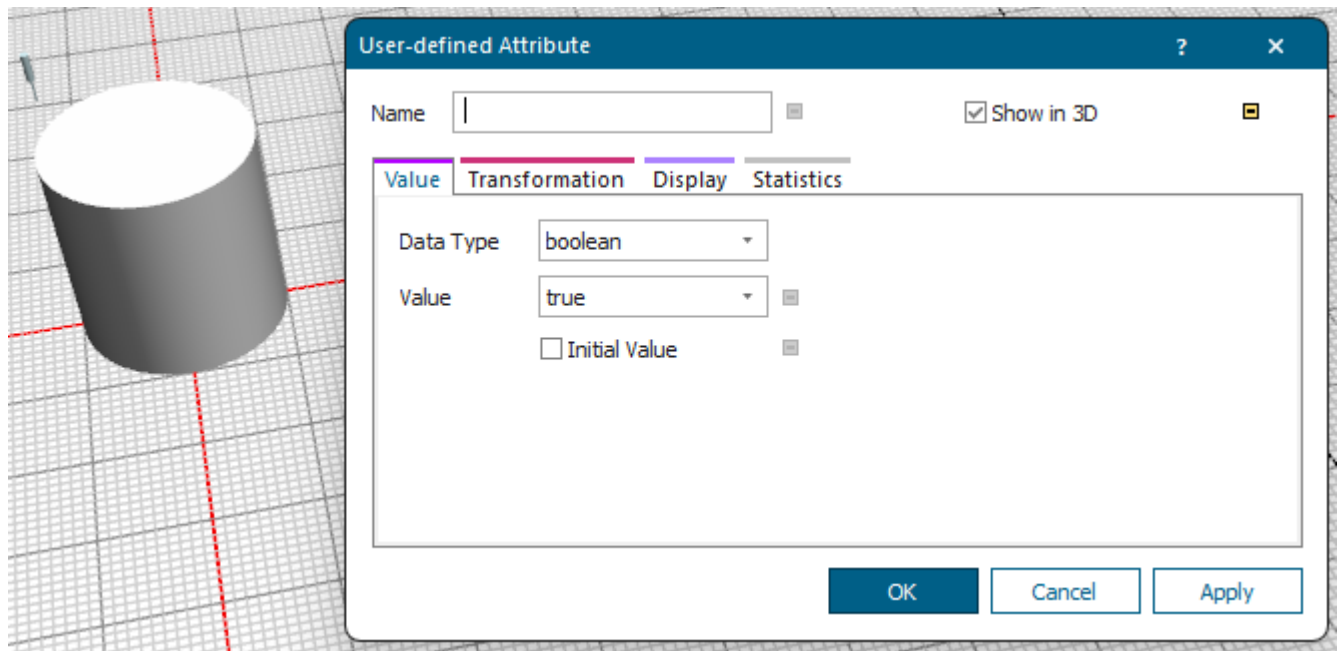
Note

This does not apply to a *Frame* or a folder.



Plant Simulation opens the dialog **User-defined Attribute** with the **Transformation** and **Display Settings** that the dialog had when you last created a *user-defined attribute*. **Show in 3D** is selected by default.

Enter and select the settings you need and click **OK** to create the *user-defined attribute*.



Previous Scene

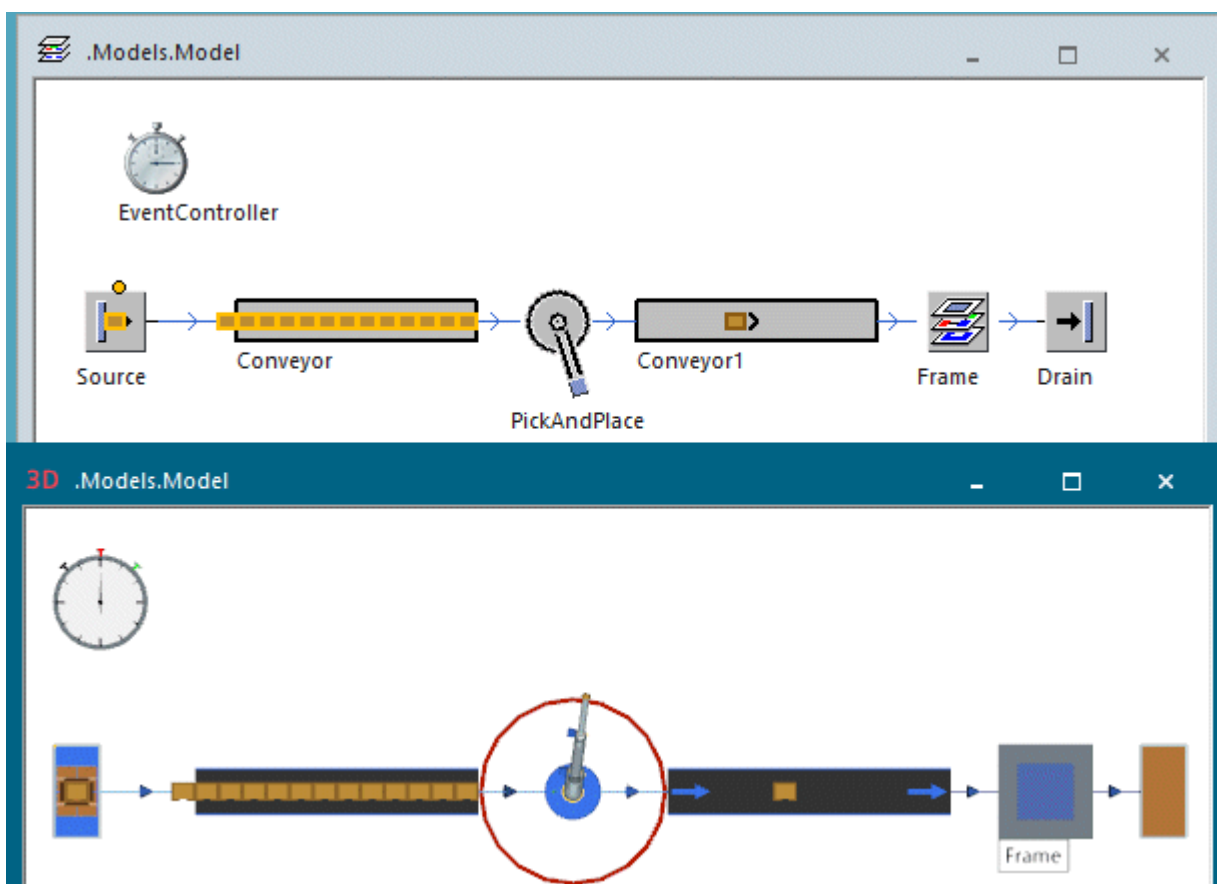
To return to the level in the hierarchy of objects or the camera mark that was active before the current level, select this command.

Video on YouTube

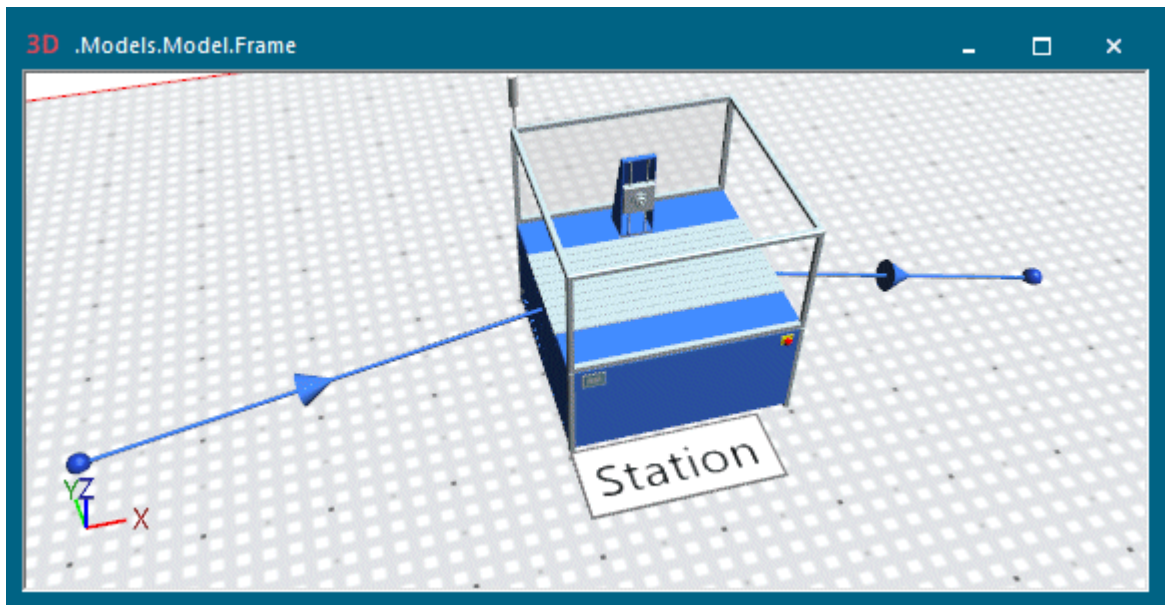
<https://youtu.be/FNpiAaUK8iM?si=dYxqljwYquKYheWo&t=571>

Adapt the 3D Model to the 2D Model

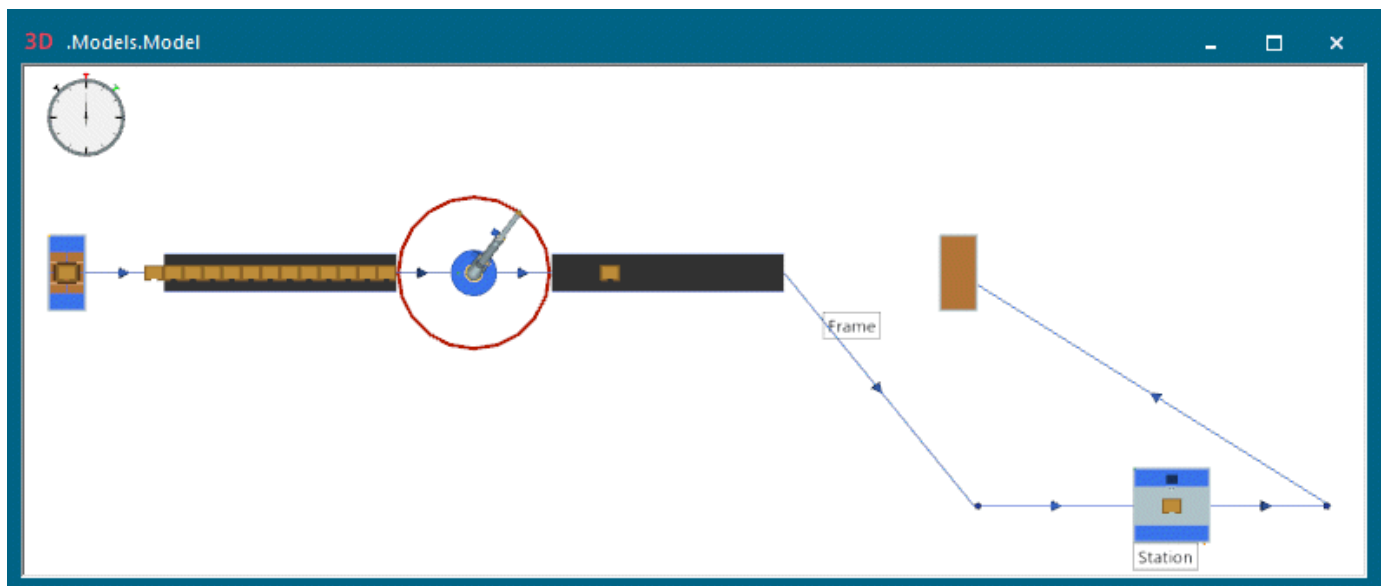
When you created your simulation model in 2D in a **2D and 3D model** while 3D was active and when your model contains a *Frame* containing objects, you might be surprised how the model looks with the standard settings in 3D.



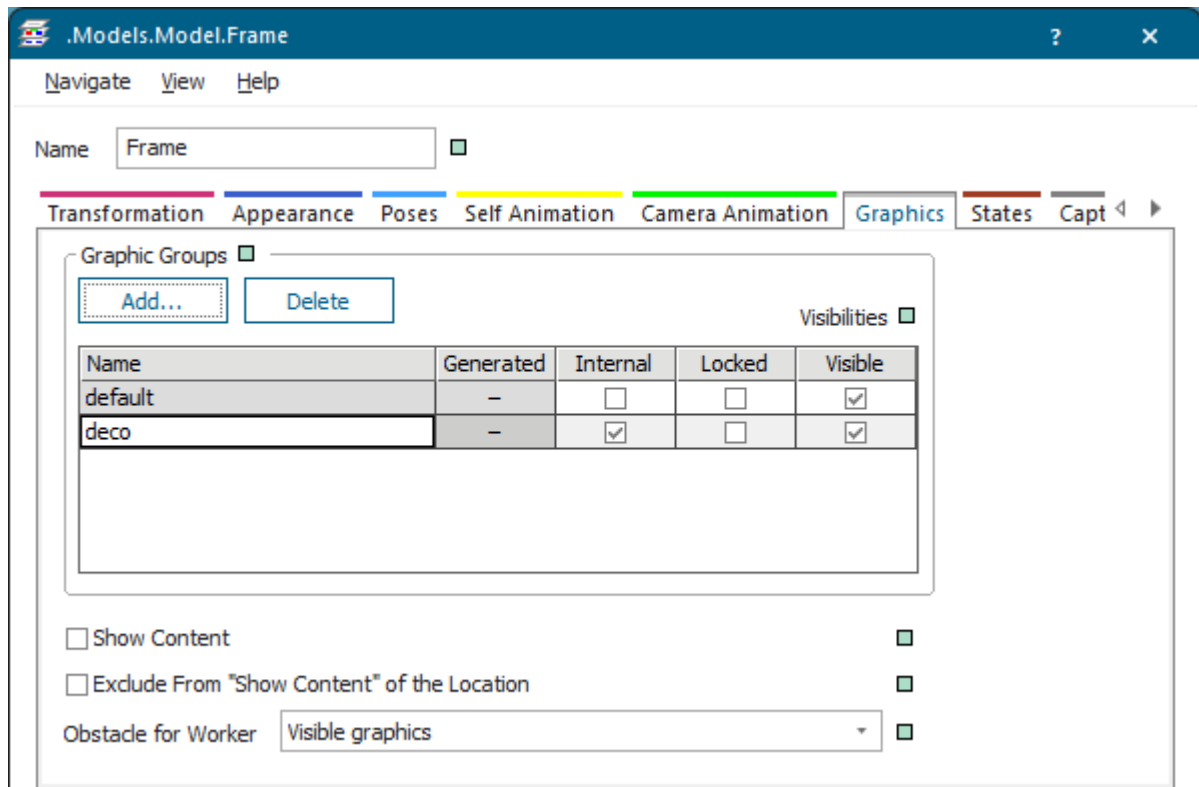
In our sample model the *Frame* contains two *Interfaces* and a *Station*.



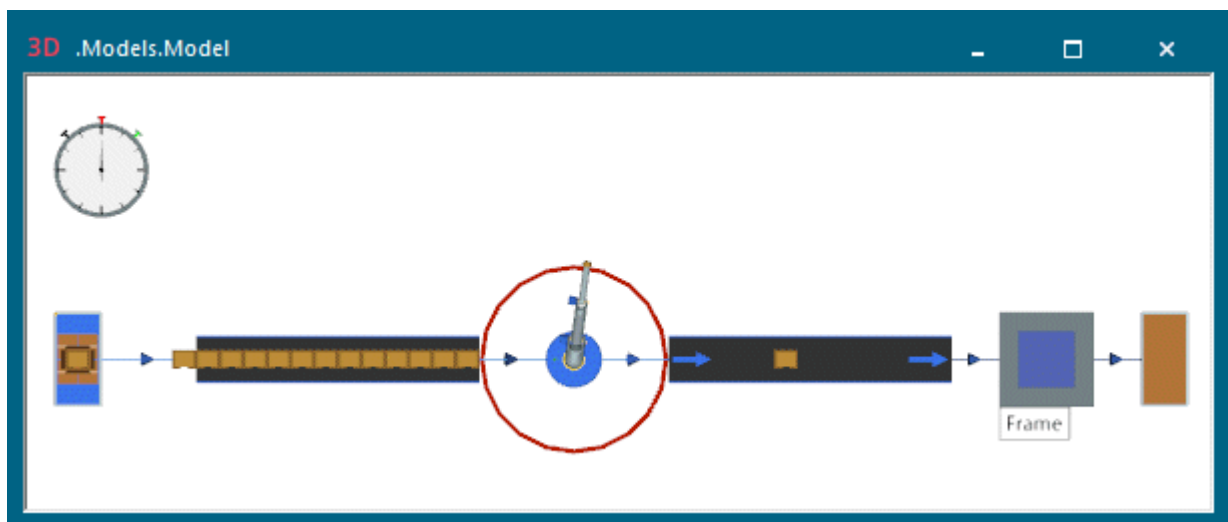
If **Show Content** is active, the content of the *Frame*, i.e., the *Interfaces* and the *Station*, are shown on the same layer as the other material flow objects. This might look strange at first sight, compare the first screenshot.



To make the 3D model look like the 2D model, we clear the check box **Show Content**.



Our model then looks like this and matches what we expected to see:



Compare [Set How Plant Simulation Shows an Object](#)

Back to [Work with Objects](#)